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A1 policy functions for open radio access network (O-RAN) systems

Abstract

The present invention relates to an apparatus comprising: memory to store policy statement information for a plurality of radio access network (RAN) automation applications (rApps); and processing circuitry, coupled with the memory, to: retrieve the policy statement information from the memory, wherein the policy statement information includes respective policy scope identifiers for respective rApps in the plurality of rApps; identify a conflict associated with common or overlapping policy scope identifiers between two or more rApps from the plurality of rApps; modify one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict; and notify the two or more rApps of the modification of the one or more A1 policies.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application is a national phase entry under 35 U.S.C. § 371 of International Application No. PCT/US2022/036130, filed Jul. 5, 2022, entitled "A1 POLICY FUNCTIONS FOR OPEN RADIO ACCESS NETWORK (O-RAN) SYSTEMS," which claims priority to U.S. Provisional Patent Application No. 63/218,861, which was filed Jul. 6, 2021, the entire disclosures of which are hereby incorporated by reference.

FIELD

(1) Various embodiments generally may relate to the field of wireless communications. For example, some embodiments may relate to features associated with A1 policy functions for open radio access network (O-RAN) systems.

BACKGROUND

(2) Among other things, the Open Radio Access Network (O-RAN) architecture helps enable intelligent radio access network (RAN) operation and optimization using artificial intelligence (AI) and machine learning (ML) in wireless communication networks. RAN intelligence controllers (RIC) are developed to manage AI/ML-assisted solutions for RAN functions. In the overall O-RAN architecture, A1 policies enable the Non-RT (Non-Real-Time) RIC to guide its connected Near-RT (Near-Real-Time) RICs to fulfil the RAN intent via the A1 interface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments will be readily understood by the following detailed description in conjunction with the accompanying drawings. To facilitate this description, like reference numerals designate like structural elements. Embodiments are illustrated by way of example and not by way of limitation in the figures of the accompanying drawings.

(2) FIG. 1 illustrates an example of a non-RT RIC architecture with A1 policy function in accordance with various embodiments.

(3) FIG. 2 illustrates an example of a procedure for RAN intent subscription in accordance with various embodiments.

(4) FIG. 3 illustrates an example of rApp querying RAN intent in a Non-RT RIC framework in accordance with various embodiments.

(5) FIG. 4 illustrates an example of an rApp subscription notification of the creation, modification, and deletion of RAN intents in accordance with various embodiments.

(6) FIG. 5 illustrates an example of a Non-RT RIC notifying an rApp of a subscribed event for RAN intent in accordance with various embodiments.

(7) FIG. 6 illustrates an example of an rApp unsubscribing its subscribed events for RAN intent in accordance with various embodiments.

(8) FIG. 7 illustrates an example of a procedure for an rApp querying supported A1 policy types in

accordance with various embodiments.

(9) FIG. 8 illustrates an example of an rApp querying supported A1 policy types in accordance with various embodiments.

(10) FIG. 9 illustrates an example of in accordance with various embodiments.

(11) FIG. 10 illustrates an example of an rApp subscription notification on supported A1 policy types (e.g., adding types, remove types) in accordance with various embodiments.

(12) FIG. 11 illustrates an example of an rApp unsubscribing from its subscribed notification in accordance with various embodiments.

(13) FIG. 12 illustrates an example of a procedure for an rApp to create an A1 policy in accordance with various embodiments.

(14) FIG. 13 illustrates an example of a procedure for an rApp to subscribe for notification of A1 policy enforcement status changes in accordance with various embodiments.

(15) FIG. 14 illustrates an example of a procedure for an rApp to update an A1 policy in accordance with various embodiments.

(16) FIG. 15 illustrates an example of a procedure for an rApp to remove an A1 policy in accordance with various embodiments.

(17) FIG. 16 illustrates an example of an rApp creating an A1 policy in accordance with various embodiments.

(18) FIG. 17 illustrates an example of an rApp updating an A1 policy in accordance with various embodiments.

(19) FIG. 18 illustrates an example of an rApp deleting an A1 policy in accordance with various embodiments.

(20) FIG. 19 illustrates an example of an rApp querying A1 policies in accordance with various embodiments.

(21) FIG. 20 illustrates an example of an rApp subscribing for notifications (e.g., new policy created, policy update or removal, policy enforcement status change) in accordance with various embodiments.

(22) FIG. 21 illustrates an example of a Non-RT RIC framework notifying subscribed events to an rApp in accordance with various embodiments.

(23) FIG. 22 illustrates an example of an rApp unsubscribing from subscribed events for A1 policies in accordance with various embodiments.

(24) FIG. 23 schematically illustrates a wireless network in accordance with various embodiments.

(25) FIG. 24 schematically illustrates components of a wireless network in accordance with various embodiments.

(26) FIG. 25 is a block diagram illustrating components, according to some example embodiments, able to read instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium) and perform any one or more of the methodologies discussed herein.

(27) FIG. 26 provides an example of a high-level view of an Open RAN (O-RAN) architecture in accordance with various embodiments.

(28) FIG. 27 illustrates an example of a Uu interface in accordance with various embodiments.

(29) FIGS. 28, 29, and 30 depict examples of procedures for practicing the various embodiments discussed herein.

DETAILED DESCRIPTION

(30) The following detailed description refers to the accompanying drawings. The same reference numbers may be used in different drawings to identify the same or similar elements. In the following description, for purposes of explanation and not limitation, specific details are set forth such as particular structures, architectures, interfaces, techniques, etc. in order to provide a thorough understanding of the various aspects of various embodiments. However, it will be apparent to those skilled in the art having the benefit of the present disclosure that the various

aspects of the various embodiments may be practiced in other examples that depart from these specific details. In certain instances, descriptions of well-known devices, circuits, and methods are omitted so as not to obscure the description of the various embodiments with unnecessary detail. For the purposes of the present document, the phrases “A or B” and “A/B” mean (A), (B), or (A and B).

(31) Various embodiments herein provide techniques for A1 policy functions in the Non-RT RIC framework to produce A1-P related services in the R1 interface. In some embodiments, an A1 policy function hosts the A1 policy repository, which stores A1 policies generated by an rApp, also referred to as a RAN automation application. Among other things, rApps may help optimize network traffic. Additionally, rApps can subscribe to notifications of A1 policy updates and supported A1 policy types of a specific policy scope identifier.

(32) Some embodiments provide an A1 policy function in the Non-RT RIC. In some embodiments the A1 policy function may host an A1 policy repository and track A1 policy enforcement status. The A1 policy function may also track supported A1 policy types in connected Near-RT RICs. Among other things, embodiments of the present disclosure provide solutions for the Non-RT RIC architecture (including A1 policy functions) and related procedures, which have not been specified yet in O-RAN WG2.

(33) A1 Policy

(34) An A1 policy is a declarative policy, e.g., it expresses the goals of the policy, but it does not restrict any implementations on how the goals are achieved. In O-RAN, an A1 policy is used to enable Non-RT RIC/SMO to guide Near-RT RIC towards the fulfilment of the RAN intent.

(35) An A1 policy includes a scope identifier and one or more policy statements. The scope identifier specifies the subjects of the policy statements, e.g., UEs and cells. The policy statement represents the policy goals, and it can contain policy objectives and policy resources. Policy objectives are statements with objectives to be reached in the policy. Policy resources are statements specifying usage of RAN resources.

(36) An A1 policy may be created, modified, and deleted by the Non-RT RIC. Until a policy's deletion, it should be enforced in the Near-RT RIC. If the enforcement status changes, the Near-RT RIC would inform the Non-RT RIC.

(37) A1 Policy Function

(38) FIG. 1 illustrates an example of a non-RT RIC architecture with A1 policy function.

(39) In one embodiment, the A1 policy function creates, updates, and deletes A1 policies in the Non-RT RIC. A1 policy function subscribes data analytics provided by rApps via interfacing with data collection and repository functions.

(40) In another embodiment, policy generation rApps create, update, and delete A1 policies in the Non-RT RIC: Based on the injected RAN intent, the rApp creates A1 policies. Based on the network observation over the O1 interface, the rApp decides whether the RAN intent is fulfilled or not. If the RAN intent is not fulfilled, the rApp modifies/updates the A1 policies. If the RAN intent is not fulfilled, the rApp can delete the A1 policies.

(41) The A1 policy function provides the following functionalities: It hosts RAN intent repository. Policy generation rApp can subscribe/unsubscribe notifications of the creation, modification, and deletion of RAN intents. It hosts A1 policy repository. Policy generation rApp store generated policy in the policy repository, and A1 policy function assign an ID to each stored policy. Note that this ID is not necessary as same as the Policy Id used in the A1 spec. Service consumer rApp can query A1 policies with specific policy scope identifiers (e.g., A1 policies applied to a specific UE or specific cell, etc.) or Near-RT RIC (e.g., A1 policies in a specific Near-RT RIC) Service consumer rApp can subscribe/unsubscribe notifications of the creation, modification, and deletion of A1 policies with specific policy scope identifiers (e.g., A1 policies applied to a specific UE or specific cell, etc.) or Near-RT RIC (e.g., A1 policies in a specific Near-RT RIC). It performs conflict mitigation. If A1 policy generation rApps set contradicting policy statements to the same or

overlapping policy scope identifiers, the A1 policy function resolves the conflicts by modifying one or more A1 policies. A1 policy function notifies the change of A1 policies to their generation rApp. Examples of conflicting policy statements are Examples for policy objective conflicts: same UE with different priority Level in QoS optimization, same slice with different qosScore in QoS optimization, etc. Example for policy resource conflicts: same UE with different preference on the same cell in traffic steering policy, etc. It tracks the enforcement status of all A1 policies created in the Non-RT RIC. Service consumer rApp can subscribe the enforcement status of A1 policies. A1 policy function notifies rApp when the enforcement status of subscribed A1 policy changes. It tracks the supported A1 policy types of each connected Near-RT RIC. In one embodiment, policy generation rApp can request and subscribe/unsubscribe supported policy types given specific policy identifier. A1 policy function identify the right Near-RT RIC(s) based on the policy identifier. A1 policy function notifies supported policy types of subscribed policy identifier, based on the supported policy types of identified Near-RT RIC(s). In another embodiment, policy generation rApp request and subscribe/unsubscribe supported policy types of Near-RT RICs. A1 policy function notify policy generation rApp when the supported policy types of subscribed Near-RT RIC changes. It creates, updates, and deletes A1 policies in the Near-RT RIC via A1 termination and A1 interface. It queries A1 policy status in the Near-RT RIC, and it receives A1 policy feedback from Near-RT RIC This functionality may also include related south-bound services toward A1 termination.

A1-P Related Services in the R1 Interface

(42) In one embodiment, Table 1 shows the A1-P related service in the R1 interface, produced by the A1-policy function.

(43) TABLE-US-00001 TABLE 1 A1-P related services in the R1 interface produced by the A1-policy function

Operation	Service Name	Service operation semantics	Description
R1_RanIntentDiscovery	RanIntentQuery	Request/response	Policy generation rApp requests RAN intent in the A1 policy function
RanIntentSubscribe	Subscribe/notify	Policy generation rApp subscribes to be notified, if interested RAN intents is created, updated, and removed.	
RanIntentNotify	A1 policy function notifies	Policy generation rApp of its subscribed events on RAN intents	
RanIntentUnsubscribe	Policy generation rApp unsubscribes	its subscribed events on RAN intents	
R1_A1PolicyManagement	A1PolicyCreate	Request/response	Policy generation rApp creates and stores a generated A1 policy in the A1 policy function, based on the RAN intent. The A1 policy function responds with an ID for created policy
A1PolicyUpdate	Request/response	Policy generation rApp modifies an A1 policy stored in the A1 policy function, based on the network observation over O1 interface.	
A1PolicyDelete	Request/response	Policy generation rApp removes an A1 policy stored in the A1 policy function	
A1PolicyQuery	Request/response	rApp requests A1 policies in the A1 policy function with specific scope identifier or Near-RT RIC	
A1PolicySubscribe	Subscribe/notify	rApp subscribes to be notified, if A1 policies applied to specific scope identifier are created, updated, removed, and if their enforcement status changes.	
A1PolicyNotify	A1 policy function notifies	rApp of its subscribed events on A1 policies	
A1PolicyUnsubscribe	rApp unsubscribes	its subscribed events on A1 policies	
R1_A1PolicyTypeDiscovery	A1PolicyTypeQuery	Request/response	Policy generation rApp requests supported A1 policy types with specific scope identifier or Near-RT RIC
A1PolicyTypeSubscribe	Subscribe/notify	rApp subscribes to be notified, if supported A1 policy types of specific scope identifier or Near-RT RIC is changed.	
A1PolicyTypeNotify	A1 policy function notifies	rApp of its subscribed events on supported A1 policy types.	
A1PolicyTypeUnsubscribe	rApp unsubscribes	its subscribed events on supported A1 policy types.	

A1 Policy Related Procedures

RAN Intent Subscription

(44) An example of a procedure for an rApp subscribing for RAN intent in A1 policy functions is illustrated in FIG. 2. Step 1: The A1 policy generation rApp send subscription request to the Non-

RT RIC framework. Step 2: The subscription request is forwarded to the A1 policy function through R1 termination. Step 3: A1 policy function sends back subscription response. Step 4: Subscription response is forwarded to the policy generation rApp through R1 termination. Step 5: RAN intent is injected into the Non-RT RIC framework. Step 6: RAN intent is forwarded to RAN intent repository in the A1 policy function through external termination. Step 7: A1 policy function sends notification about the newly injected RAN intent. Step 8: The notification is forwarded to policy generation rApp through R1 termination.

(45) For an R1 interface, examples of the service operations of a R1_RanIntentDiscovery service are illustrated in FIGS. 3 through 6.

(46) Supported A1 Policy Type Query

(47) An example of a procedure for an rApp querying supported A1 policy types is illustrated in FIG. 7: Step 1: rApp sends query request about supported A1 policy types to the Non-RT RIC. Policy generation rApp needs to know what types A1 policy is supported in the Near-RT RIC before generating the A1 policy. The query can be filtered by scope identifiers (e.g., UE id, cell id, etc.) or the query can be made toward a specific Near-RT RIC. Step 2: Query request is forwarded to A1 policy function through R1 termination Step 3: A1 policy function figure out supported A1 policy types Step 4: A1 policy function sends back query response with a list of supported A1 policy types. Step 5: Query response is forwarded to rApp through R1 termination

(48) For the R1 interface, examples of the service operations of R1_A1PolicyTypeDiscovery service are illustrated in FIGS. 8-11.

(49) A1 Policy Creation

(50) An example of a procedure for creating an A1 policy is illustrated in FIG. 12: Step 1: Policy generation rApp generates A1 policy based on the RAN intent. Step 2: Policy generation rApp sends A1 policy creation request to Non-RT RIC framework Step 3: A1 policy creation request is forwarded to A1 policy function through R1 termination Step 4: A1 policy function creates A1 policy in the Near-RT RIC via A1 interface Step 5: A1 policy function sends back A1 policy creation response with an identification for created policy. Note that this identification is not same as A1 policy ID in A1 specification. Step 6: A1 policy creation response is forwarded to rApp through R1 termination

A1 Policy Status Subscription

(51) An example of a procedure for an rApp subscribing for notification of an A1 policy enforcement status change is illustrated in FIG. 13: Step 1: Policy generation rApp subscribes notification of A1 policy, if its enforcement status changes. Step 2: Subscription request is forwarded to A1 policy function through R1 termination Step 3: A1 policy function sends back subscription response. Step 4: Subscription response is forwarded to the rApp through R1 termination. Step 5, 6: A1 policy feedback is sent by the Near-RT RIC to indicate the change of A1 policy enforcement status to A1 policy function through A1 termination Step 7: A1 policy function notifies rApp about the change of the subscribed A1 policy enforcement status Step 8: Notification is forwarded to policy generation rApp through R1 termination.

A1 Policy Update

(52) An example of a procedure for an rApp updating an A1 policy is illustrated in FIG. 14: Step 1: Policy generation rApp subscribes O1 data from data collection and repository function for network performance observation. Step 2: Policy generation rApp updates A1 policy based on RAN intent and network performance. Step 3: Policy generation rApp sends A1 policy modification request to the Non-RT RIC framework Step 4: Modification request is forwarded to A1 policy function through R1 termination Step 5: A1 policy function update A1 policy in the Near-RT RIC via A1 interface Step 6: A1 policy function sends back policy modification response. Step 7: Policy modification response is forwarded to the rApp through R1 termination.

A1 Policy Deletion

(53) An example of a procedure for an rApp deleting an A1 policy is illustrated in FIG. 15: Step 1:

Policy generation rApp sends A1 policy deletion request to the Non-RT RIC. Step 2: Deletion request is forwarded to A1 policy function. Step 3: A1 policy function remove A1 policy in the Near-RT RIC via A1 interface. Step 4: A1 policy function send back policy deletion response Step 5: Policy deletion response is forwarded to the policy generation rApp through R1 termination. (54) For an R1 interface, examples of the service operations of an R1_A1PolicyManagement service are illustrated in FIGS. **16-22**.

(55) Systems and Implementations

(56) FIGS. **23-27** illustrate various systems, devices, and components that may implement aspects of disclosed embodiments.

(57) FIG. **23** illustrates a network **2300** in accordance with various embodiments. The network **2300** may operate in a manner consistent with 3GPP technical specifications for LTE or 5G/NR systems. However, the example embodiments are not limited in this regard and the described embodiments may apply to other networks that benefit from the principles described herein, such as future 3GPP systems, or the like.

(58) The network **2300** may include a UE **2302**, which may include any mobile or non-mobile computing device designed to communicate with a RAN **2304** via an over-the-air connection. The UE **2302** may be communicatively coupled with the RAN **2304** by a Uu interface. The UE **2302** may be, but is not limited to, a smartphone, tablet computer, wearable computer device, desktop computer, laptop computer, in-vehicle infotainment, in-car entertainment device, instrument cluster, head-up display device, onboard diagnostic device, dashtop mobile equipment, mobile data terminal, electronic engine management system, electronic/engine control unit, electronic/engine control module, embedded system, sensor, microcontroller, control module, engine management system, networked appliance, machine-type communication device, M2M or D2D device, IoT device, etc.

(59) In some embodiments, the network **2300** may include a plurality of UEs coupled directly with one another via a sidelink interface. The UEs may be M2M/D2D devices that communicate using physical sidelink channels such as, but not limited to, PSBCH, PSDCH, PSSCH, PSCCH, PSFCH, etc.

(60) In some embodiments, the UE **2302** may additionally communicate with an AP **2306** via an over-the-air connection. The AP **2306** may manage a WLAN connection, which may serve to offload some/all network traffic from the RAN **2304**. The connection between the UE **2302** and the AP **2306** may be consistent with any IEEE 802.11 protocol, wherein the AP **2306** could be a wireless fidelity (Wi-Fi®) router. In some embodiments, the UE **2302**, RAN **2304**, and AP **2306** may utilize cellular-WLAN aggregation (for example, LWA/LWIP). Cellular-WLAN aggregation may involve the UE **2302** being configured by the RAN **2304** to utilize both cellular radio resources and WLAN resources.

(61) The RAN **2304** may include one or more access nodes, for example, AN **2308**. AN **2308** may terminate air-interface protocols for the UE **2302** by providing access stratum protocols including RRC, PDCP, RLC, MAC, and L1 protocols. In this manner, the AN **2308** may enable data/voice connectivity between CN **2320** and the UE **2302**. In some embodiments, the AN **2308** may be implemented in a discrete device or as one or more software entities running on server computers as part of, for example, a virtual network, which may be referred to as a CRAN or virtual baseband unit pool. The AN **2308** be referred to as a BS, gNB, RAN node, eNB, ng-eNB, NodeB, RSU, TRxP, TRP, etc. The AN **2308** may be a macrocell base station or a low power base station for providing femtocells, picocells or other like cells having smaller coverage areas, smaller user capacity, or higher bandwidth compared to macrocells.

(62) In embodiments in which the RAN **2304** includes a plurality of ANs, they may be coupled with one another via an X2 interface (if the RAN **2304** is an LTE RAN) or an Xn interface (if the RAN **2304** is a 5G RAN). The X2/Xn interfaces, which may be separated into control/user plane interfaces in some embodiments, may allow the ANs to communicate information related to

handovers, data/context transfers, mobility, load management, interference coordination, etc.

(63) The ANs of the RAN **2304** may each manage one or more cells, cell groups, component carriers, etc. to provide the UE **2302** with an air interface for network access. The UE **2302** may be simultaneously connected with a plurality of cells provided by the same or different ANs of the RAN **2304**. For example, the UE **2302** and RAN **2304** may use carrier aggregation to allow the UE **2302** to connect with a plurality of component carriers, each corresponding to a Pcell or Scell. In dual connectivity scenarios, a first AN may be a master node that provides an MCG and a second AN may be secondary node that provides an SCG. The first/second ANs may be any combination of eNB, gNB, ng-eNB, etc.

(64) The RAN **2304** may provide the air interface over a licensed spectrum or an unlicensed spectrum. To operate in the unlicensed spectrum, the nodes may use LAA, eLAA, and/or feLAA mechanisms based on CA technology with PCells/Scells. Prior to accessing the unlicensed spectrum, the nodes may perform medium/carrier-sensing operations based on, for example, a listen-before-talk (LBT) protocol.

(65) In V2X scenarios the UE **2302** or AN **2308** may be or act as a RSU, which may refer to any transportation infrastructure entity used for V2X communications. An RSU may be implemented in or by a suitable AN or a stationary (or relatively stationary) UE. An RSU implemented in or by a UE may be referred to as a “UE-type RSU”: an eNB may be referred to as an “eNB-type RSU”: a gNB may be referred to as a “gNB-type RSU”: and the like. In one example, an RSU is a computing device coupled with radio frequency circuitry located on a roadside that provides connectivity support to passing vehicle UEs. The RSU may also include internal data storage circuitry to store intersection map geometry, traffic statistics, media, as well as applications/software to sense and control ongoing vehicular and pedestrian traffic. The RSU may provide very low latency communications required for high speed events, such as crash avoidance, traffic warnings, and the like. Additionally or alternatively, the RSU may provide other cellular/WLAN communications services. The components of the RSU may be packaged in a weatherproof enclosure suitable for outdoor installation, and may include a network interface controller to provide a wired connection (e.g., Ethernet) to a traffic signal controller or a backhaul network.

(66) In some embodiments, the RAN **2304** may be an LTE RAN **2310** with eNBs, for example, eNB **2312**. The LTE RAN **2310** may provide an LTE air interface with the following characteristics: SCS of 15 KHz: CP-OFDM waveform for DL and SC-FDMA waveform for UL; turbo codes for data and TBCC for control: etc. The LTE air interface may rely on CSI-RS for CSI acquisition and beam management: PDSCH/PDCCH DMRS for PDSCH/PDCCH demodulation: and CRS for cell search and initial acquisition, channel quality measurements, and channel estimation for coherent demodulation/detection at the UE. The LTE air interface may operating on sub-6 GHZ bands.

(67) In some embodiments, the RAN **2304** may be an NG-RAN **2314** with gNBs, for example, gNB **2316**, or ng-eNBs, for example, ng-eNB **2318**. The gNB **2316** may connect with 5G-enabled UEs using a 5G NR interface. The gNB **2316** may connect with a 5G core through an NG interface, which may include an N2 interface or an N3 interface. The ng-eNB **2318** may also connect with the 5G core through an NG interface, but may connect with a UE via an LTE air interface. The gNB **2316** and the ng-eNB **2318** may connect with each other over an Xn interface.

(68) In some embodiments, the NG interface may be split into two parts, an NG user plane (NG-U) interface, which carries traffic data between the nodes of the NG-RAN **2314** and a UPF **2348** (e.g., N3 interface), and an NG control plane (NG-C) interface, which is a signaling interface between the nodes of the NG-RAN **2314** and an AMF **2344** (e.g., N2 interface).

(69) The NG-RAN **2314** may provide a 5G-NR air interface with the following characteristics: variable SCS: CP-OFDM for DL. CP-OFDM and DFT-s-OFDM for UL: polar, repetition, simplex, and Reed-Muller codes for control and LDPC for data. The 5G-NR air interface may rely on CSI-

RS. PDSCH/PDCCH DMRS similar to the LTE air interface. The 5G-NR air interface **20**) may not use a CRS, but may use PBCH DMRS for PBCH demodulation: PTRS for phase tracking for PDSCH: and tracking reference signal for time tracking. The 5G-NR air interface may operating on FR1 bands that include sub-6 GHz bands or FR2 bands that include bands from 24.25 GHz to 52.6 GHz. The 5G-NR air interface may include an SSB that is an area of a downlink resource grid that includes PSS/SSS/PBCH.

(70) In some embodiments, the 5G-NR air interface may utilize BWPs for various purposes. For example, BWP can be used for dynamic adaptation of the SCS. For example, the UE **2302** can be configured with multiple BWPs where each BWP configuration has a different SCS. When a BWP change is indicated to the UE **2302**, the SCS of the transmission is changed as well. Another use case example of BWP is related to power saving. In particular, multiple BWPs can be configured for the UE **2302** with different amount of frequency resources (for example, PRBs) to support data transmission under different traffic loading scenarios. A BWP containing a smaller number of PRBs can be used for data transmission with small traffic load while allowing power saving at the UE **2302** and in some cases at the gNB **2316**. A BWP containing a larger number of PRBs can be used for scenarios with higher traffic load.

(71) The RAN **2304** is communicatively coupled to CN **2320** that includes network elements to provide various functions to support data and telecommunications services to customers/subscribers (for example, users of UE **2302**). The components of the CN **2320** may be implemented in one physical node or separate physical nodes. In some embodiments, NFV may be utilized to virtualize any or all of the functions provided by the network elements of the CN **2320** onto physical compute/storage resources in servers, switches, etc. A logical instantiation of the CN **2320** may be referred to as a network slice, and a logical instantiation of a portion of the CN **2320** may be referred to as a network sub-slice.

(72) In some embodiments, the CN **2320** may be an LTE CN **2322**, which may also be referred to as an EPC. The LTE CN **2322** may include MME **2324**, SGW **2326**, SGSN **2328**, HSS **2330**, PGW **2332**, and PCRF **2334** coupled with one another over interfaces (or “reference points”) as shown. Functions of the elements of the LTE CN **2322** may be briefly introduced as follows.

(73) The MME **2324** may implement mobility management functions to track a current location of the UE **2302** to facilitate paging, bearer activation/deactivation, handovers, gateway selection, authentication, etc.

(74) The SGW **2326** may terminate an SI interface toward the RAN and route data packets between the RAN and the LTE CN **2322**. The SGW **2326** may be a local mobility anchor point for inter-RAN node handovers and also may provide an anchor for inter-3GPP mobility. Other responsibilities may include lawful intercept, charging, and some policy enforcement.

(75) The SGSN **2328** may track a location of the UE **2302** and perform security functions and access control. In addition, the SGSN **2328** may perform inter-EPC node signaling for mobility between different RAT networks: PDN and S-GW selection as specified by MME **2324**; MME selection for handovers: etc. The S3 reference point between the MME **2324** and the SGSN **2328** may enable user and bearer information exchange for inter-3GPP access network mobility in idle/active states.

(76) The HSS **2330** may include a database for network users, including subscription-related information to support the network entities handling of communication sessions. The HSS **2330** can provide support for routing/roaming, authentication, authorization, naming/addressing resolution, location dependencies, etc. An S6a reference point between the HSS **2330**) and the MME **2324** may enable transfer of subscription and authentication data for authenticating/authorizing user access to the LTE CN **2320**.

(77) The PGW **2332** may terminate an SGi interface toward a data network (DN) **2336** that may include an application/content server **2338**. The PGW **2332** may route data packets between the LTE CN **2322** and the data network **2336**. The PGW **2332** may be coupled with the SGW **2326** by

an S5 reference point to facilitate user plane tunneling and tunnel management. The PGW **2332** may further include a node for policy enforcement and charging data collection (for example, PCEF). Additionally, the SGi reference point between the PGW **2332** and the data network **2336** may be an operator external public, a private PDN, or an intra-operator packet data network, for example, for provision of IMS services. The PGW **2332** may be coupled with a PCRF **2334** via a Gx reference point.

(78) The PCRF **2334** is the policy and charging control element of the LTE CN **2322**. The PCRF **2334** may be communicatively coupled to the app/content server **2338** to determine appropriate QoS and charging parameters for service flows. The PCRF **2332** may provision associated rules into a PCEF (via Gx reference point) with appropriate TFT and QCI.

(79) In some embodiments, the CN **2320** may be a 5GC **2340**. The 5GC **2340** may include an AUSF **2342**, AMF **2344**, SMF **2346**, UPF **2348**, NSSF **2350**, NEF **2352**, NRF **2354**, PCF **2356**, UDM **2358**, and AF **2360** coupled with one another over interfaces (or “reference points”) as shown. Functions of the elements of the 5GC **2340** may be briefly introduced as follows.

(80) The AUSF **2342** may store data for authentication of UE **2302** and handle authentication-related functionality. The AUSF **2342** may facilitate a common authentication framework for various access types. In addition to communicating with other elements of the 5GC **2340** over reference points as shown, the AUSF **2342** may exhibit an Nausf service-based interface.

(81) The AMF **2344** may allow other functions of the 5GC **2340** to communicate with the UE **2302** and the RAN **2304** and to subscribe to notifications about mobility events with respect to the UE **2302**. The AMF **2344** may be responsible for registration management (for example, for registering UE **2302**), connection management, reachability management, mobility management, lawful interception of AMF-related events, and access authentication and authorization. The AMF **2344** may provide transport for SM messages between the UE **2302** and the SMF **2346**, and act as a transparent proxy for routing SM messages. AMF **2344** may also provide transport for SMS messages between UE **2302** and an SMSF. AMF **2344** may interact with the AUSF **2342** and the UE **2302** to perform various security anchor and context management functions. Furthermore, AMF **2344** may be a termination point of a RAN CP interface, which may include or be an N2 reference point between the RAN **2304** and the AMF **2344**; and the AMF **2344** may be a termination point of NAS (N1) signaling, and perform NAS ciphering and integrity protection. AMF **2344** may also support NAS signaling with the UE **2302** over an N3 IWF interface.

(82) The SMF **2346** may be responsible for SM (for example, session establishment, tunnel management between UPF **2348** and AN **2308**); UE IP address allocation and management (including optional authorization); selection and control of UP function; configuring traffic steering at UPF **2348** to route traffic to proper destination; termination of interfaces toward policy control functions; controlling part of policy enforcement, charging, and QoS; lawful intercept (for SM events and interface to LI system); termination of SM parts of NAS messages; downlink data notification; initiating AN specific SM information, sent via AMF **2344** over N2 to AN **2308**; and determining SSC mode of a session. SM may refer to management of a PDU session, and a PDU session or “session” may refer to a PDU connectivity service that provides or enables the exchange of PDUs between the UE **2302** and the data network **2336**.

(83) The UPF **2348** may act as an anchor point for intra-RAT and inter-RAT mobility, an external PDU session point of interconnect to data network **2336**, and a branching point to support multi-homed PDU session. The UPF **2348** may also perform packet routing and forwarding, perform packet inspection, enforce the user plane part of policy rules, lawfully intercept packets (UP collection), perform traffic usage reporting, perform QoS handling for a user plane (e.g., packet filtering, gating, UL/DL rate enforcement), perform uplink traffic verification (e.g., SDF-to-QoS flow mapping), transport level packet marking in the uplink and downlink, and perform downlink packet buffering and downlink data notification triggering. UPF **2348** may include an uplink classifier to support routing traffic flows to a data network.

(84) The NSSF **2350** may select a set of network slice instances serving the UE **2302**. The NSSF **2350** may also determine allowed NSSAI and the mapping to the subscribed S-NSSAIs, if needed. The NSSF **2350** may also determine the AMF set to be used to serve the UE **2302**, or a list of candidate AMFs based on a suitable configuration and possibly by querying the NRF **2354**. The selection of a set of network slice instances for the UE **2302** may be triggered by the AMF **2344** with which the UE **2302** is registered by interacting with the NSSF **2350**, which may lead to a change of AMF. The NSSF **2350** may interact with the AMF **2344** via an N22 reference point; and may communicate with another NSSF in a visited network via an N31 reference point (not shown). Additionally, the NSSF **2350** may exhibit an Nnssf service-based interface.

(85) The NEF **2352** may securely expose services and capabilities provided by 3GPP network functions for third party, internal exposure/re-exposure. AFs (e.g., AF **2360**), edge computing or fog computing systems, etc. In such embodiments, the NEF **2352** may authenticate, authorize, or throttle the AFs. NEF **2352** may also translate information exchanged with the AF **2360** and information exchanged with internal network functions. For example, the NEF **2352** may translate between an AF-Service-Identifier and an internal 5GC information. NEF **2352** may also receive information from other NFs based on exposed capabilities of other NFs. This information may be stored at the NEF **2352** as structured data, or at a data storage NF using standardized interfaces. The stored information can then be re-exposed by the NEF **2352** to other NFs and AFs, or used for other purposes such as analytics. Additionally, the NEF **2352** may exhibit an Nnef service-based interface.

(86) The NRF **2354** may support service discovery functions, receive NF discovery requests from NF instances, and provide the information of the discovered NF instances to the NF instances. NRF **2354** also maintains information of available NF instances and their supported services. As used herein, the terms “instantiate.” “instantiation.” and the like may refer to the creation of an instance, and an “instance” may refer to a concrete occurrence of an object, which may occur, for example, during execution of program code. Additionally, the NRF **2354** may exhibit the Nnrf service-based interface.

(87) The PCF **2356** may provide policy rules to control plane functions to enforce them, and may also support unified policy framework to govern network behavior. The PCF **2356** may also implement a front end to access subscription information relevant for policy decisions in a UDR of the UDM **2358**. In addition to communicating with functions over reference points as shown, the PCF **2356** exhibit an Npcf service-based interface.

(88) The UDM **2358** may handle subscription-related information to support the network entities' handling of communication sessions, and may store subscription data of UE **2302**. For example, subscription data may be communicated via an N8 reference point between the UDM **2358** and the AMF **2344**. The UDM **2358** may include two parts, an application front end and a UDR. The UDR may store subscription data and policy data for the UDM **2358** and the PCF **2356**, and/or structured data for exposure and application data (including PFDs for application detection, application request information for multiple UEs **2302**) for the NEF **2352**. The Nudr service-based interface may be exhibited by the UDR **221** to allow the UDM **2358**, PCF **2356**, and NEF **2352** to access a particular set of the stored data, as well as to read, update (e.g., add, modify), delete, and subscribe to notification of relevant data changes in the UDR. The UDM may include a UDM-FE, which is in charge of processing credentials, location management, subscription management and so on. Several different front ends may serve the same user in different transactions. The UDM-FE accesses subscription information stored in the UDR and performs authentication credential processing, user identification handling, access authorization, registration/mobility management, and subscription management. In addition to communicating with other NFs over reference points as shown, the UDM **2358** may exhibit the Nudm service-based interface.

(89) The AF **2360** may provide application influence on traffic routing, provide access to NEF, and interact with the policy framework for policy control.

(90) In some embodiments, the 5GC **2340** may enable edge computing by selecting operator/3rd party services to be geographically close to a point that the UE **2302** is attached to the network. This may reduce latency and load on the network. To provide edge-computing implementations, the 5GC **2340** may select a UPF **2348** close to the UE **2302** and execute traffic steering from the UPF **2348** to data network **2336** via the N6 interface. This may be based on the UE subscription data, UE location, and information provided by the AF **2360**. In this way, the AF **2360** may influence UPF (re)selection and traffic routing. Based on operator deployment, when AF **2360** is considered to be a trusted entity, the network operator may permit AF **2360** to interact directly with relevant NFs. Additionally, the AF **2360** may exhibit an Naf service-based interface.

(91) The data network **2336** may represent various network operator services, Internet access, or third party services that may be provided by one or more servers including, for example, application/content server **2338**.

(92) FIG. **24** schematically illustrates a wireless network **2400** in accordance with various embodiments. The wireless network **2400** may include a UE **2402** in wireless communication with an AN **2404**. The UE **2402** and AN **2404** may be similar to, and substantially interchangeable with, like-named components described elsewhere herein.

(93) The UE **2402** may be communicatively coupled with the AN **2404** via connection **2406**. The connection **2406** is illustrated as an air interface to enable communicative coupling, and can be consistent with cellular communications protocols such as an LTE protocol or a 5G NR protocol operating at mmWave or sub-6 GHZ frequencies.

(94) The UE **2402** may include a host platform **2408** coupled with a modem platform **2410**. The host platform **2408** may include application processing circuitry **2412**, which may be coupled with protocol processing circuitry **2414** of the modem platform **2410**. The application processing circuitry **2412** may run various applications for the UE **2402** that source/sink application data. The application processing circuitry **2412** may further implement one or more layer operations to transmit/receive application data to/from a data network. These layer operations may include transport (for example UDP) and Internet (for example, IP) operations

(95) The protocol processing circuitry **2414** may implement one or more of layer operations to facilitate transmission or reception of data over the connection **2406**. The layer operations implemented by the protocol processing circuitry **2414** may include, for example, MAC, RLC, PDCP, RRC and NAS operations.

(96) The modem platform **2410** may further include digital baseband circuitry **2416** that may implement one or more layer operations that are “below” layer operations performed by the protocol processing circuitry **2414** in a network protocol stack. These operations may include, for example, PHY operations including one or more of HARQ-ACK functions, scrambling/descrambling, encoding/decoding, layer mapping/de-mapping, modulation symbol mapping, received symbol/bit metric determination, multi-antenna port precoding/decoding, which may include one or more of space-time, space-frequency or spatial coding, reference signal generation/detection, preamble sequence generation and/or decoding, synchronization sequence generation/detection, control channel signal blind decoding, and other related functions.

(97) The modem platform **2410** may further include transmit circuitry **2418**, receive circuitry **2420**, RF circuitry **2422**, and RF front end (RFFE) **2424**, which may include or connect to one or more antenna panels **2426**. Briefly, the transmit circuitry **2418** may include a digital-to-analog converter, mixer, intermediate frequency (IF) components, etc.; the receive circuitry **2420** may include an analog-to-digital converter, mixer, IF components, etc.; the RF circuitry **2422** may include a low-noise amplifier, a power amplifier, power tracking components, etc.; RFFE **2424** may include filters (for example, surface/bulk acoustic wave filters), switches, antenna tuners, beamforming components (for example, phase-array antenna components), etc. The selection and arrangement of the components of the transmit circuitry **2418**, receive circuitry **2420**, RF circuitry **2422**, RFFE **2424**, and antenna panels **2426** (referred generically as “transmit/receive components”) may be

specific to details of a specific implementation such as, for example, whether communication is TDM or FDM, in mmWave or sub-6 GHz frequencies, etc. In some embodiments, the transmit/receive components may be arranged in multiple parallel transmit/receive chains, may be disposed in the same or different chips/modules, etc.

(98) In some embodiments, the protocol processing circuitry **2414** may include one or more instances of control circuitry (not shown) to provide control functions for the transmit/receive components.

(99) A UE reception may be established by and via the antenna panels **2426**, RFFE **2424**, RF circuitry **2422**, receive circuitry **2420**, digital baseband circuitry **2416**, and protocol processing circuitry **2414**. In some embodiments, the antenna panels **2426** may receive a transmission from the AN **2404** by receive-beamforming signals received by a plurality of antennas/antenna elements of the one or more antenna panels **2426**.

(100) A UE transmission may be established by and via the protocol processing circuitry **2414**, digital baseband circuitry **2416**, transmit circuitry **2418**, RF circuitry **2422**, RFFE **2424**, and antenna panels **2426**. In some embodiments, the transmit components of the UE **2404** may apply a spatial filter to the data to be transmitted to form a transmit beam emitted by the antenna elements of the antenna panels **2426**.

(101) Similar to the UE **2402**, the AN **2404** may include a host platform **2428** coupled with a modem platform **2430**. The host platform **2428** may include application processing circuitry **2432** coupled with protocol processing circuitry **2434** of the modem platform **2430**. The modem platform may further include digital baseband circuitry **2436**, transmit circuitry **2438**, receive circuitry **2440**, RF circuitry **2442**, RFFE circuitry **2444**, and antenna panels **2446**. The components of the AN **2404** may be similar to and substantially interchangeable with like-named components of the UE **2402**. In addition to performing data transmission/reception as described above, the components of the AN **2408** may perform various logical functions that include, for example, RNC functions such as radio bearer management, uplink and downlink dynamic radio resource management, and data packet scheduling.

(102) FIG. **25** is a block diagram illustrating components, according to some example embodiments, able to read instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. **25** shows a diagrammatic representation of hardware resources **2500** including one or more processors (or processor cores) **2510**, one or more memory/storage devices **2520**, and one or more communication resources **2530**, each of which may be communicatively coupled via a bus **2540** or other interface circuitry. For embodiments where node virtualization (e.g., NFV) is utilized, a hypervisor **2502** may be executed to provide an execution environment for one or more network slices/sub-slices to utilize the hardware resources **2500**.

(103) The processors **2510** may include, for example, a processor **2512** and a processor **2514**. The processors **2510** may be, for example, a central processing unit (CPU), a reduced instruction set computing (RISC) processor, a complex instruction set computing (CISC) processor, a graphics processing unit (GPU), a DSP such as a baseband processor, an ASIC, an FPGA, a radio-frequency integrated circuit (RFIC), another processor (including those discussed herein), or any suitable combination thereof.

(104) The memory/storage devices **2520** may include main memory, disk storage, or any suitable combination thereof. The memory/storage devices **2520** may include, but are not limited to, any type of volatile, non-volatile, or semi-volatile memory such as dynamic random access memory (DRAM), static random access memory (SRAM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), Flash memory, solid-state storage, etc.

(105) The communication resources **2530** may include interconnection or network interface

controllers, components, or other suitable devices to communicate with one or more peripheral devices **2504** or one or more databases **2506** or other network elements via a network **2508**. For example, the communication resources **2530** may include wired communication components (e.g., for coupling via USB, Ethernet, etc.), cellular communication components, NFC components, Bluetooth® (or Bluetooth® Low Energy) components, Wi-Fi® components, and other communication components.

(106) Instructions **2550** may comprise software, a program, an application, an applet, an app, or other executable code for causing at least any of the processors **2510** to perform any one or more of the methodologies discussed herein. The instructions **2550** may reside, completely or partially, within at least one of the processors **2510** (e.g., within the processor's cache memory), the memory/storage devices **2520**, or any suitable combination thereof. Furthermore, any portion of the instructions **2550** may be transferred to the hardware resources **2500** from any combination of the peripheral devices **2504** or the databases **2506**. Accordingly, the memory of processors **2510**, the memory/storage devices **2520**, the peripheral devices **2504**, and the databases **2506** are examples of computer-readable and machine-readable media.

(107) FIG. **26** provides a high-level view of an Open RAN (O-RAN) architecture **2600**. The O-RAN architecture **2600** includes four O-RAN defined interfaces—namely, the A1 interface, the O1 interface, the O2 interface, and the Open Fronthaul Management (M)-plane interface—which connect the Service Management and Orchestration (SMO) framework **2602** to O-RAN network functions (NFs) **2604** and the O-Cloud **2606**. The SMO **2602** also connects with an external system **2610**, which provides data to the SMO **2602**. FIG. **26** also illustrates that the A1 interface terminates at an O-RAN Non-Real Time (RT) RAN Intelligent Controller (RIC) **2612** in or at the SMO **2602** and at the O-RAN Near-RT RIC **2614** in or at the O-RAN NFs **2604**. The O-RAN NFs **2604** can be VNFs such as VMs or containers, sitting above the O-Cloud **2606** and/or Physical Network Functions (PNFs) utilizing customized hardware. All O-RAN NFs **2604** are expected to support the O1 interface when interfacing the SMO framework **2602**. The O-RAN NFs **2604** connect to the NG-Core **2608** via the NG interface (which is a 3GPP defined interface). The Open Fronthaul M-plane interface between the SMO **2602** and the O-RAN Radio Unit (O-RU) **2616** supports the O-RU **2616** management in the O-RAN hybrid model. The Open Fronthaul M-plane **20** interface is an optional interface to the SMO **2602** that is included for backward compatibility purposes, and is intended for management of the O-RU **2616** in hybrid mode only. The management architecture of flat mode and its relation to the O1 interface for the O-RU **2616** is for future study. The O-RU **2616** termination of the O1 interface towards the SMO **2602**.

(108) FIG. **27** shows an O-RAN logical architecture **2700** corresponding to the O-RAN architecture **2600** of FIG. **26**. In FIG. **27**, the SMO **2702** corresponds to the SMO **2602**. O-Cloud **2706** corresponds to the O-Cloud **2606**, the non-RT RIC **2712** corresponds to the non-RT RIC **2612**, the near-RT RIC **2714** corresponds to the near-RT RIC **2614**, and the O-RU **2716** corresponds to the O-RU **2616** of FIG. **27**, respectively. The O-RAN logical architecture **2700** includes a radio portion and a management portion.

(109) The management portion/side of the architectures **2700** includes the SMO Framework **2702** containing the non-RT RIC **2712**, and may include the O-Cloud **2706**. The O-Cloud **2706** is a cloud computing platform including a collection of physical infrastructure nodes to host the relevant O-RAN functions (e.g., the near-RT RIC **2714**. O-CU-CP **2721**. O-CU-UP **2722**, and the O-DU **2715**), supporting software components (e.g., OSs. VMMs, container runtime engines, ML engines, etc.), and appropriate management and orchestration functions.

(110) The radio portion/side of the logical architecture **2700** includes the near-RT RIC **2714**, the O-RAN Distributed Unit (O-DU) **2715**, the O-RU **2716**, the O-RAN Central Unit-Control Plane (O-CU-CP) **2721**, and the O-RAN Central Unit-User Plane (O-CU-UP) **2722** functions. The radio portion/side of the logical architecture **2700** may also include the O-e/gNB **2710**.

(111) The O-DU **2715** is a logical node hosting RLC. MAC, and higher PHY layer

entities/elements (High-PHY layers) based on a lower layer functional split. The O-RU 2716 is a logical node hosting lower PHY layer entities/elements (Low-PHY layer) (e.g., FFT/iFFT, PRACH extraction, etc.) and RF processing elements based on a lower layer functional split. Virtualization of O-RU 2716 is FFS. The O-CU-CP 2721 is a logical node hosting the RRC and the control plane (CP) part of the PDCP protocol. The O O-CU-UP 2722 is a logical node hosting the user plane part of the PDCP protocol and the SDAP protocol.

(112) An E2 interface terminates at a plurality of E2 nodes. The E2 nodes are logical nodes/entities that terminate the E2 interface. For NR/5G access, the E2 nodes include the O-CU-CP 2721, O-CU-UP 2722, O-DU 2715, or any combination of elements. For E-UTRA access the E2 nodes include the O-e/gNB 2710. As shown in FIG. 27, the E2 interface also connects the O-e/gNB 2710 to the Near-RT RIC 2714. The protocols over E2 interface are based exclusively on Control Plane (CP) protocols. The E2 functions are grouped into the following categories: (a) near-RT RIC 2714 services (REPORT, INSERT, CONTROL and POLICY); and (b) near-RT RIC 2714 support functions, which include E2 Interface Management (E2 Setup, E2 Reset, Reporting of General Error Situations, etc.) and Near-RT RIC Service Update (e.g., capability exchange related to the list of E2 Node functions exposed over E2).

(113) FIG. 27 shows the Uu interface between a UE 2701 and O-e/gNB 2710 as well as between the UE 2701 and O-RAN components. The Uu interface is a 3GPP defined interface, which includes a complete protocol stack from L1 to L3 and terminates in the NG-RAN or E-UTRAN. The O-e/gNB 2710 is an LTE eNB, a 5G gNB or ng-eNB that supports the E2 interface. The O-e/gNB 2710 may be the same or similar as AN 2308 and/or AN 2404 discussed previously. The UE 2701 may correspond to UE 2302 and/or UE 2402 discussed with respect to FIGS. 23 and 24, and/or the like. There may be multiple UEs 2701 and/or multiple O-e/gNB 2710, each of which may be connected to one another the via respective Uu interfaces. Although not shown in FIG. 27, the O-e/gNB 2710 supports O-DU 2715 and O-RU 2716 functions with an Open Fronthaul interface between them.

(114) The Open Fronthaul (OF) interface(s) is/are between O-DU 2715 and O-RU 2716 functions. The OF interface(s) includes the Control User Synchronization (CUS) Plane and Management (M) Plane. FIGS. 26 and 27 also show that the O-RU 2716 terminates the OF M-Plane interface towards the O-DU 2715 and optionally towards the SMO 2702. The O-RU 2716 terminates the OF CUS-Plane interface towards the O-DU 2715 and the SMO 2702.

(115) The F1-c interface connects the O-CU-CP 2721 with the O-DU 2715. As defined by 3GPP, the F1-c interface is between the gNB-CU-CP and gNB-DU nodes. However, for purposes of O-RAN, the F1-c interface is adopted between the O-CU-CP 2721 with the O-DU 2715 functions while reusing the principles and protocol stack defined by 3GPP and the definition of interoperability profile specifications.

(116) The F1-u interface connects the O-CU-UP 2722 with the O-DU 2715. As defined by 3GPP, the F1-u interface is between the gNB-CU-UP and gNB-DU nodes. However, for purposes of O-RAN, the F1-u interface is adopted between the O-CU-UP 2722 with the O-DU 2715 functions while reusing the principles and protocol stack defined by 3GPP and the definition of interoperability profile specifications.

(117) The NG-c interface is defined by 3GPP as an interface between the gNB-CU-CP and the AMF in the 5GC. The NG-c is also referred as the N2 interface. The NG-u interface is defined by 3GPP, as an interface between the gNB-CU-UP and the UPF in the 5GC. The NG-u interface is referred as the N3 interface. In O-RAN, NG-c and NG-u protocol stacks defined by 3GPP are reused and may be adapted for O-RAN purposes.

(118) The X2-c interface is defined in 3GPP for transmitting control plane information between eNBs or between eNB and en-gNB in EN-DC. The X2-u interface is defined in 3GPP for transmitting user plane information between eNBs or between eNB and en-gNB in EN-DC. In O-RAN, X2-c and X2-u protocol stacks defined by 3GPP are reused and may be adapted for O-RAN

purposes

(119) The Xn-c interface is defined in 3GPP for transmitting control plane information between gNBs, ng-eNBs, or between an ng-eNB and gNB. The Xn-u interface is defined in 3GPP for transmitting user plane information between gNBs, ng-eNBs, or between ng-eNB and gNB. In O-RAN, Xn-c and Xn-u protocol stacks defined by 3GPP are reused and may be adapted for O-RAN purposes

(120) The E1 interface is defined by 3GPP as being an interface between the gNB-CU-CP and gNB-CU-UP. In O-RAN, E1 protocol stacks defined by 3GPP are reused and adapted as being an interface between the O-CU-CP **2721** and the O-CU-UP **2722** functions.

(121) The O-RAN Non-Real Time (RT) RAN Intelligent Controller (RIC) **2712** is a logical function within the SMO framework **2602**, **2702** that enables non-real-time control and optimization of RAN elements and resources: AI/machine learning (ML) workflow(s) including model training, inferences, and updates; and policy-based guidance of applications/features in the Near-RT RIC **2714**.

(122) The O-RAN near-RT RIC **2714** is a logical function that enables near-real-time control and optimization of RAN elements and resources via fine-grained data collection and actions over the E2 interface. The near-RT RIC **2714** may include one or more AI/ML workflows including model training, inferences, and updates.

(123) The non-RT RIC **2712** can be an ML training host to host the training of one or more ML models. ML training can be performed offline using data collected from the RIC. O-DU **2715** and O-RU **2716**. For supervised learning, non-RT RIC **2712** is part of the SMO **2702**, and the ML training host and/or ML model host/actor can be part of the non-RT RIC **2712** and/or the near-RT RIC **2714**. For unsupervised learning, the ML training host and ML model host/actor can be part of the non-RT RIC **2712** and/or the near-RT RIC **2714**. For reinforcement learning, the ML training host and ML model host/actor may be co-located as part of the non-RT RIC **2712** and/or the near-RT RIC **2714**. In some implementations, the non-RT RIC **2712** may request or trigger ML model training in the training hosts regardless of where the model is deployed and executed. ML models may be trained and not currently deployed.

(124) In some implementations, the non-RT RIC **2712** provides a query-able catalog for an ML designer/developer to publish/install trained ML models (e.g., executable software components). In these implementations, the non-RT RIC **2712** may provide discovery mechanism if a particular ML model can be executed in a target ML inference host (MF), and what number and type of ML models can be executed in the MF. For example, there may be three types of ML catalogs made discoverable by the non-RT RIC **2712**: a design-time catalog (e.g., residing outside the non-RT RIC **2712** and hosted by some other ML platform(s)), a training/deployment-time catalog (e.g., residing inside the non-RT RIC **2712**), and a run-time catalog (e.g., residing inside the non-RT RIC **2712**). The non-RT RIC **2712** supports necessary capabilities for ML model inference in support of ML assisted solutions running in the non-RT RIC **2712** or some other ML inference host. These capabilities enable executable software to be installed such as VMs, containers, etc. The non-RT RIC **2712** may also include and/or operate one or more ML engines, which are packaged software executable libraries that provide methods, routines, data types, etc., used to run ML models. The non-RT RIC **2712** may also implement policies to switch and activate ML model instances under different operating conditions.

(125) The non-RT RIC **272** is be able to access feedback data (e.g., FM and PM statistics) over the O1 interface on ML model performance and perform necessary evaluations. If the ML model fails during runtime, an alarm can be generated as feedback to the non-RT RIC **2712**. How well the ML model is performing in terms of prediction accuracy or other operating statistics it produces can also be sent to the non-RT RIC **2712** over O1. The non-RT RIC **2712** can also scale ML model instances running in a target MF over the O1 interface by observing resource utilization in MF. The environment where the ML model instance is running (e.g., the MF) monitors resource utilization

of the running ML model. This can be done, for example, using an ORAN-SC component called ResourceMonitor in the near-RT RIC **2714** and/or in the non-RT RIC **2712**, which continuously monitors resource utilization. If resources are low or fall below a certain threshold, the runtime environment in the near-RT RIC **2714** and/or the non-RT RIC **2712** provides a scaling mechanism to add more ML instances. The scaling mechanism may include a scaling factor such as an number, percentage, and/or other like data used to scale up/down the number of ML instances. ML model instances running in the target ML inference hosts may be automatically scaled by observing resource utilization in the MF. For example, the Kubernetes® (K8s) runtime environment typically provides an auto-scaling feature.

(126) The A1 interface is between the non-RT RIC **2712** (within or outside the SMO **2702**) and the near-RT RIC **2714**. The A1 interface supports three types of services, including a Policy Management Service, an Enrichment Information Service, and ML Model Management Service. A1 policies have the following characteristics compared to persistent configuration: A1 policies are not critical to traffic: A1 policies have temporary validity: A1 policies may handle individual UE or dynamically defined groups of UEs: A1 policies act within and take precedence over the configuration: and A1 policies are non-persistent, e.g., do not survive a restart of the near-RT RIC.

EXAMPLE PROCEDURES

(127) In some embodiments, the electronic device(s), network(s), system(s), chip(s) or component(s), or portions or implementations thereof, of FIGS. **10-12**, or some other figure herein, may be configured to perform one or more processes, techniques, or methods as described herein, or portions thereof. One such process is depicted in FIG. **28**. In this example, process **2800** includes, at **2805**, retrieving, from a memory, policy statement information for a plurality of radio access network (RAN) automation applications (rApps), wherein the policy statement information includes respective policy scope identifiers for respective rApps in the plurality of rApps. The process further includes, at **2810**, identifying a conflict associated with common or overlapping policy scope identifiers between two or more rApps from the plurality of rApps. The process further includes, at **2815**, modifying one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict. The process further includes, at **2820**, notifying the two or more rApps of the modification of the one or more A1 policies.

(128) Another such example is illustrated in FIG. **29**. In this example, process **2900** includes, at **2905**, identifying a conflict associated with common or overlapping policy scope identifiers between two or more radio access network (RAN) automation applications (rApps) from a plurality of rApps based on policy statement information that includes respective policy scope identifiers for respective rApps in the plurality of rApps. The process further includes, at **2910**, modifying one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict. The process further includes, at **2915**, notifying the two or more rApps of the modification of the one or more A1 policies.

(129) Another such example is illustrated in FIG. **30**. In this example, process **3000** includes, at **3005**, Identifying a conflict associated with common or overlapping policy scope identifiers between two or more radio access network (RAN) automation applications (rApps) from a plurality of rApps based on policy statement information that includes respective policy scope identifiers for respective rApps in the plurality of rApps. The process further includes, at **3010**, modifying one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict. The process further includes, at **3015**, notifying the two or more rApps of the modification of the one or more A1 policies. The process further includes, at **3020**, tracking a supported A1 policy type for each of a plurality of near-RT RICs, and either: notifying an rApp of a supported A1 policy type for a near-RT RIC based on a policy identifier, or notifying an rApp of a change in supported policy

types for a near-RT RIC

(130) For one or more embodiments, at least one of the components set forth in one or more of the preceding figures may be configured to perform one or more operations, techniques, processes, and/or methods as set forth in the example section below. For example, the baseband circuitry as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below. For another example, circuitry associated with a UE, base station, network element, etc. as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below in the example section.

EXAMPLES

(131) Example 1 may include a method of an A1 policy function to provide one or more of the following functionalities in the Non-RT RIC framework: It hosts RAN intent repository. A1 policy function stores injected RAN intent. It hosts A1 policy repository. A1 policy function stores A1 policy created by policy generation rApps. A1 policy function response to rApp's query about A1 policies. A1 policy function notifies rApps about their subscribed events on A1 policies. It performs conflict mitigation for A1 policies. If A1 policies created by policy generation rApps set contradicting policy statements to the same or overlapping policy scope identifiers, the A1 policy function resolves the conflicts by modifying one or more A1 policies. A1 policy function notifies the change of A1 policies to their generation rApp. It tracks the enforcement status of all A1 policies created in the Non-RT RIC. A1 policy function notifies rApp when the enforcement status of subscribed A1 policy changes. It tracks the supported A1 policy types of each connected Near-RT RIC.

(132) Example 2 may include the method of example 1 or some other example herein, wherein A1 policy function provides one or more of the following services to rApps: RAN intent discovery. A1 policy management Supported A1 policy management discovery

(133) Example 3 may include rApp can query RAN intent available in the Non-RT RIC framework.

(134) Example 4 may include rApp can subscribe, unsubscribe RAN intent from A1 policy function. A1 policy function notifies rApp of its subscribed events on RAN intent, e.g., New RAN intent available; RAN intent is modified: and/or RAN intent is deleted, etc.

(135) Example 5 may include rApp can generate A1 policy based on RAN intent and send it to the A1 policy function (via POST method). A1 policy function assigns an identifier for generated A1 policy in the response. The identifier is not the same of the A1 policy ID used in A1 specification

(136) Example 6 rApp can modify A1 policy based on RAN intent and network performance observation (O1 data). rApp send the updated A1 policy to the A1 policy function (e.g., via PUT method).

(137) Example 7 may include rApp can delete A1 policy (e.g., via DELETE method).

(138) Example 8 may include rApp can query A1 policy (e.g., via GET method) by providing filtering on the scope identifier (e.g., UE id, Cell id, etc.). A1 policy function returns a list of A1 policy ID, which satisfies the filtering in the query request.

(139) Example 9 may include rApp can subscribe (e.g., via POST method), unsubscribe (e.g., via DELETE method) notifications about A1 policy from A1 policy function. A1 policy function notifies rApp of its subscribed events on A1 policies, e.g., New A1 policy is created; A1 policy is modified; A1 policy is deleted; and/or The enforcement status of A1 policy is changed, etc.

(140) Example 10 may include rApp can query supported A1 policy (e.g., via GET method) by providing filtering on the scope identifier (e.g., UE id, Cell id, Near-RT RIC id, etc.). A1 policy function returns a list of supported A1 policy type ID.

(141) Example 11 may include rApp can subscribe (e.g., via POST method), unsubscribe (e.g., via DELETE method) notifications about supported A1 policy types from A1 policy function. A1 policy function notifies rApp of its subscribed events on supported A1 policy types, e.g., New supported A1 policy types are added Some of supported A1 policy types are removed, etc.

(142) Example 12 may include A1 policy related procedures are proposed for RAN intent subscription, rApp querying supported A1 policy types, A1 policy generation, rApp subscribing policy enforcement status feedback, A1 policy update, A1 deletion.

(143) Example X1 includes an apparatus comprising: memory to store policy statement information for a plurality of radio access network (RAN) automation applications (rApps); and processing circuitry, coupled with the memory, to: retrieve the policy statement information from the memory, wherein the policy statement information includes respective policy scope identifiers for respective rApps in the plurality of rApps; identify a conflict associated with common or overlapping policy scope identifiers between two or more rApps from the plurality of rApps; modify one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict; and notify the two or more rApps of the modification of the one or more A1 policies.

(144) Example X2 includes the apparatus of example X1 or some other example herein, wherein the processing circuitry is further to: track an enforcement status of an A1 policy created in the non-RT RIC; and notify an rApp subscribed to the tracked A1 policy when the enforcement status of the tracked A1 policy changes.

(145) Example X3 includes the apparatus of example X1 or some other example herein, wherein the processing circuitry is further to track a supported A1 policy type for each of a plurality of near-RT RICs.

(146) Example X4 includes the apparatus of example X3 or some other example herein, wherein the processing circuitry is to notify an rApp of a supported A1 policy type for a near-RT RIC based on a policy identifier.

(147) Example X5 includes the apparatus of example X3 or some other example herein, wherein the processing circuitry is to notify an rApp of a change in supported policy types for a near-RT RIC.

(148) Example X6 includes the apparatus of example X1 or some other example herein, wherein the processing circuitry is further to generate an A1-related service that includes: a RAN intent discovery service, an A1 policy management service, or an A1 policy type discovery service.

(149) Example X7 includes the apparatus of example X1 or some other example herein, wherein the processing circuitry is further to: receive a supported A1 policy types query request via an R1 termination; and in response to the supported A1 policy types query request, send a query response that includes an indication of supported A1 policy types to a policy generation rApp via the R1 termination.

(150) Example X8 includes the apparatus of example X1 or some other example herein, wherein the processing circuitry is further to: receive an A1 policy feedback message from the near-RT RIC via an A1 termination to indicate a change of A1 policy enforcement status; and send a notification of the A1 policy enforcement status change to a policy generation rApp via an R1 termination.

(151) Example X9 includes the apparatus of example X1 or some other example herein, wherein the processing circuitry is further to: receive an A1 policy modification request from a policy generation rApp via an R1 termination; update an A1 policy over the A1 interface based on the A1 policy modification request; and send an A1 policy modification response to the policy generation rApp via the R1 termination.

(152) Example X10 includes the apparatus of example X1 or some other example herein, wherein the processing circuitry is further to: receive an A1 policy deletion request from a policy generation rApp via an R1 termination; remove an A1 policy over the A1 interface based on the A1 policy deletion request; and send an A1 policy deletion response to the policy generation rApp via the R1 termination.

(153) Example X11 includes the apparatus of any of examples X1-X10 or some other example herein, wherein the processing circuitry is to implement an A1 policy function.

(154) Example X12 includes one or more computer-readable media storing instructions that, when

executed by one or more processors, cause an A1 policy function to: identify a conflict associated with common or overlapping policy scope identifiers between two or more radio access network (RAN) automation applications (rApps) from a plurality of rApps based on policy statement information that includes respective policy scope identifiers for respective rApps in the plurality of rApps; modify one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict; and notify the two or more rApps of the modification of the one or more A1 policies.

(155) Example X13 includes the one or more computer-readable media of example X12 or some other example herein, wherein the media further stores instructions to: track an enforcement status of an A1 policy created in the non-RT RIC; and notify an rApp subscribed to the tracked A1 policy when the enforcement status of the tracked A1 policy changes.

(156) Example X14 includes the one or more computer-readable media of example X12 or some other example herein, wherein the media further stores instructions to: track a supported A1 policy type for each of a plurality of near-RT RICs

(157) Example X15 includes the one or more computer-readable media of example X14 or some other example herein, wherein the media further stores instructions to: notify an rApp of a supported A1 policy type for a near-RT RIC based on a policy identifier; or notify an rApp of a change in supported policy types for a near-RT RIC.

(158) Example X16 includes the one or more computer-readable media of example X12 or some other example herein, wherein the media further stores instructions to: receive a supported A1 policy types query request via an R1 termination; and in response to the supported A1 policy types query request, send a query response that includes an indication of supported A1 policy types to a policy generation rApp via the R1 termination.

(159) Example X17 includes the one or more computer-readable media of example X12 or some other example herein, wherein the media further stores instructions to: receive an A1 policy feedback message from the near-RT RIC via an A1 termination to indicate a change of A1 policy enforcement status; and send a notification of the A1 policy enforcement status change to a policy generation rApp via an R1 termination.

(160) Example X18 includes the one or more computer-readable media of example X12 or some other example herein, wherein the media further stores instructions to: receive an A1 policy modification request from a policy generation rApp via an R1 termination; update an A1 policy over the A1 interface based on the A1 policy modification request; and send an A1 policy modification response to the policy generation rApp via the R1 termination.

(161) Example X19 includes the one or more computer-readable media of example X12 or some other example herein, wherein the media further stores instructions to: receive an A1 policy deletion request from a policy generation rApp via an R1 termination; remove an A1 policy over the A1 interface based on the A1 policy deletion request; and send an A1 policy deletion response to the policy generation rApp via the R1 termination.

(162) Example X20 includes the one or more computer-readable media of any of Examples X12-X19 or some other example herein, wherein the media further stores instructions to generate an A1-related service that includes: a RAN intent discovery service, an A1 policy management service, or an A1 policy type discovery service.

(163) Example X21 includes one or more computer-readable media storing instructions that, when executed by one or more processors, cause an A1 policy function to: identify a conflict associated with common or overlapping policy scope identifiers between two or more radio access network (RAN) automation applications (rApps) from a plurality of rApps based on policy statement information that includes respective policy scope identifiers for respective rApps in the plurality of rApps; modify one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict; notify the two or more rApps of the modification of the one or more A1 policies; and track

a supported A1 policy type for each of a plurality of near-RT RICs, and either: notify an rApp of a supported A1 policy type for a near-RT RIC based on a policy identifier, or notify an rApp of a change in supported policy types for a near-RT RIC.

(164) Example X22 includes the one or more computer-readable media of example X21 or some other example herein, wherein the media further stores instructions to send, to an rApp via an R1 termination, one or more of; an indication of a change in an enforcement status of a tracked A1 policy; an indication of supported A1 policy types; and a notification of an A1 policy enforcement status change.

(165) Example X23 includes the one or more computer-readable media of example X21 or some other example herein, wherein the media further stores instructions to send, to: update an A1 policy over the A1 interface based on an A1 policy modification request; or remove an A1 policy over the A1 interface based on an A1 policy deletion request.

(166) Example X24 includes the one or more computer-readable media of any of examples X21-X23 or some other example herein, wherein the media further stores instructions to generate an A1-related service that includes: a RAN intent discovery service, an A1 policy management service, or an A1 policy type discovery service.

(167) Example Z01 may include an apparatus comprising means to perform one or more elements of a method described in or related to any of examples 1-X24, or any other method or process described herein.

(168) Example Z02 may include one or more non-transitory computer-readable media comprising instructions to cause an electronic device, upon execution of the instructions by one or more processors of the electronic device, to perform one or more elements of a method described in or related to any of examples 1-X24, or any other method or process described herein.

(169) Example Z03 may include an apparatus comprising logic, modules, or circuitry to perform one or more elements of a method described in or related to any of examples 1-X24, or any other method or process described herein.

(170) Example Z04 may include a method, technique, or process as described in or related to any of examples 1-X24, or portions or parts thereof.

(171) Example Z05 may include an apparatus comprising: one or more processors and one or more computer-readable media comprising instructions that, when executed by the one or more processors, cause the one or more processors to perform the method, techniques, or process as described in or related to any of examples 1-X24, or portions thereof.

(172) Example Z06 may include a signal as described in or related to any of examples 1-X24, or portions or parts thereof.

(173) Example Z07 may include a datagram, packet, frame, segment, protocol data unit (PDU), or message as described in or related to any of examples 1-X24, or portions or parts thereof, or otherwise described in the present disclosure.

(174) Example Z08 may include a signal encoded with data as described in or related to any of examples 1-X24, or portions or parts thereof, or otherwise described in the present disclosure.

(175) Example Z09 may include a signal encoded with a datagram, packet, frame, segment, protocol data unit (PDU), or message as described in or related to any of examples 1-X24, or portions or parts thereof, or otherwise described in the present disclosure.

(176) Example Z10 may include an electromagnetic signal carrying computer-readable instructions, wherein execution of the computer-readable instructions by one or more processors is to cause the one or more processors to perform the method, techniques, or process as described in or related to any of examples 1-X24, or portions thereof.

(177) Example Z11 may include a computer program comprising instructions, wherein execution of the program by a processing element is to cause the processing element to carry out the method, techniques, or process as described in or related to any of examples 1-X24, or portions thereof.

(178) Example Z12 may include a signal in a wireless network as shown and described herein.

(179) Example Z13 may include a method of communicating in a wireless network as shown and described herein.

(180) Example Z14 may include a system for providing wireless communication as shown and described herein.

(181) Example Z15 may include a device for providing wireless communication as shown and described herein.

(182) Any of the above-described examples may be combined with any other example (or combination of examples), unless explicitly stated otherwise. The foregoing description of one or more implementations provides illustration and description, but is not intended to be exhaustive or to limit the scope of embodiments to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of various embodiments.

(183) Abbreviations

(184) Unless used differently herein, terms, definitions, and abbreviations may be consistent with terms, definitions, and abbreviations defined in 3GPP TR 21.905 v16.0.0 (2019-06). For the purposes of the present document, the following abbreviations may apply to the examples and embodiments discussed herein.

(185) TABLE-US-00002 3GPP Third Generation Partnership Project 4G Fourth Generation 5G Fifth Generation 5GC 5G Core network AC Application Client ACR Application Context Relocation ACK Acknowledgement ACID Application Client Identification AF Application Function AM Acknowledged Mode AMBR Aggregate Maximum Bit Rate AMF Access and Mobility Management Function AN Access Network ANR Automatic Neighbour Relation AOA Angle of Arrival AP Application Protocol, Antenna Port, Access Point API Application Programming Interface APN Access Point Name ARP Allocation and Retention Priority ARQ Automatic Repeat Request AS Access Stratum ASP Application Service Provider ASN.1 Abstract Syntax Notation One AUSF Authentication Server Function AWGN Additive White Gaussian Noise BAP Backhaul Adaptation Protocol BCH Broadcast Channel BER Bit Error Ratio BFD Beam Failure Detection BLER Block Error Rate BPSK Binary Phase Shift Keying BRAS Broadband Remote Access Server BSS Business Support System BS Base Station BSR Buffer Status Report BW Bandwidth BWP Bandwidth Part C-RNTI Cell Radio Network Temporary Identity CA Carrier Aggregation, Certification Authority CAPEX CAPital EXpenditure CBRA Contention Based Random Access CC Component Carrier, Country Code, Cryptographic Checksum CCA Clear Channel Assessment CCE Control Channel Element CCCH Common Control Channel CE Coverage Enhancement CDM Content Delivery Network CDMA Code-Division Multiple Access CDR Charging Data Request CDR Charging Data Response CFRA Contention Free Random Access CG Cell Group CGF Charging Gateway Function CHF Charging Function CI Cell Identity CID Cell-ID (e.g., positioning method) CIM Common Information Model CIR Carrier to Interference Ratio CK Cipher Key CM Connection Management, Conditional Mandatory CMAS Commercial Mobile Alert Service CMD Command CMS Cloud Management System CO Conditional Optional CoMP Coordinated Multi-Point CORESET Control Resource Set COTS Commercial Off-The-Shelf CP Control Plane, Cyclic Prefix, Connection Point CPD Connection Point Descriptor CPE Customer Premise Equipment CPICH Common Pilot Channel CQI Channel Quality Indicator CPU CSI processing unit, Central Processing Unit C/R Command/Response field bit CRAN Cloud Radio Access Network, Cloud RAN CRB Common Resource Block CRC Cyclic Redundancy Check CRI Channel-State Information Resource Indicator, CSI-RS Resource Indicator C-RNTI Cell RNTI CS Circuit Switched CSCF call session control function CSAR Cloud Service Archive CSI Channel-State Information CSI-IM CSI Interference Measurement CSI-RS CSI Reference Signal CSI-RSRP CSI reference signal received power CSI-RSRQ CSI reference signal received quality CSI-SINR CSI signal-to-noise and interference ratio CSMA Carrier Sense Multiple Access CSMA/CA CSMA with collision avoidance CSS Common Search Space, Cell-specific Search Space CTF Charging Trigger Function

CTS Clear-to-Send CW Codeword CWS Contention Window Size D2D Device-to-Device DC Dual Connectivity, Direct Current DCI Downlink Control Information DF Deployment Flavour DL Downlink DMTF Distributed Management Task Force DPDK Data Plane Development Kit DM-RS, DMRS Demodulation Reference Signal DN Data network DNN Data Network Name DNAI Data Network Access Identifier DRB Data Radio Bearer DRS Discovery Reference Signal DRX Discontinuous Reception DSL Domain Specific Language. Digital Subscriber Line DSLAM DSL Access Multiplexer DwPTS Downlink Pilot Time Slot E-LAN Ethernet Local Area Network E2E End-to-End EAS Edge Application Server ECCA extended clear channel assessment, extended CCA ECCE Enhanced Control Channel Element, Enhanced CCE ED Energy Detection EDGE Enhanced Datarates for GSM Evolution (GSM Evolution) EAS Edge Application Server EASID Edge Application Server Identification ECS Edge Configuration Server ECSP Edge Computing Service Provider EDN Edge Data Network EEC Edge Enabler Client EECID Edge Enabler Client Identification EES Edge Enabler Server EESID Edge Enabler Server Identification EHE Edge Hosting Environment EGMF Exposure Governance Management Function EGPRS Enhanced GPRS EIR Equipment Identity Register eLAA enhanced Licensed Assisted Access, enhanced LAA EM Element Manager eMBB Enhanced Mobile Broadband EMS Element Management System eNB evolved NodeB, E-UTRAN Node B EN-DC E-UTRA-NR Dual Connectivity EPC Evolved Packet Core EPDCCH enhanced PDCCH, enhanced Physical Downlink Control Channel EPRE Energy per resource element EPS Evolved Packet System EREG enhanced REG, enhanced resource element groups ETSI European Telecommunications Standards Institute ETWS Earthquake and Tsunami Warning System eUICC embedded UICC, embedded Universal Integrated Circuit Card E-UTRA Evolved UTRA E-UTRAN Evolved UTRAN EV2X Enhanced V2X F1AP F1 Application Protocol F1-C F1 Control plane interface F1-U F1 User plane interface FACCH Fast Associated Control Channel FACCH/F Fast Associated Control Channel/Full rate FACCH/H Fast Associated Control Channel/Half rate FACH Forward Access Channel FAUSCH Fast Uplink Signalling Channel FB Functional Block FBI Feedback Information FCC Federal Communications Commission FCCH Frequency Correction Channel FDD Frequency Division Duplex FDM Frequency Division Multiplex FDMA Frequency Division Multiple Access FE Front End FEC Forward Error Correction FFS For Further Study FFT Fast Fourier Transformation feLAA further enhanced Licensed Assisted Access, further enhanced LAA FN Frame Number FPGA Field-Programmable Gate Array FR Frequency Range FQDN Fully Qualified Domain Name G-RNTI GERAN Radio Network Temporary Identity GERAN GSM EDGE RAN, GSM EDGE Radio Access Network GGSN Gateway GPRS Support Node GLONASS GLObal'naya NAvigatsionnaya Sputnikovaya Sistema (Engl.: Global Navigation Satellite System) gNB Next Generation NodeB gNB-CU gNB-centralized unit, Next Generation NodeB centralized unit gNB-DU gNB-distributed unit, Next Generation NodeB distributed unit GNSS Global Navigation Satellite System GPRS General Packet Radio Service GPSI Generic Public Subscription Identifier GSM Global System for Mobile Communications, Groupe Spécial Mobile GTP GPRS Tunneling Protocol GTP-U GPRS Tunnelling Protocol for User Plane GTS Go To Sleep Signal (related to WUS) GUMMEI Globally Unique MME Identifier GUTI Globally Unique Temporary UE Identity HARQ Hybrid ARQ, Hybrid Automatic Repeat Request HANDO Handover HFN HyperFrame Number HHO Hard Handover HLR Home Location Register HN Home Network HO Handover HPLMN Home Public Land Mobile Network HSDPA High Speed Downlink Packet Access HSN Hopping Sequence Number HSPA High Speed Packet Access HSS Home Subscriber Server HSUPA High Speed Uplink Packet Access HTTP Hyper Text Transfer Protocol HTTPS Hyper Text Transfer Protocol Secure (https is http/1.1 over SSL, i.e. port 443) I-Block Information Block ICCID Integrated Circuit Card Identification IAB Integrated Access and Backhaul ICIC Inter-Cell Interference Coordination ID Identity, identifier IDFT Inverse Discrete Fourier Transform IE Information element IBE In-Band Emission IEEE Institute of Electrical and Electronics Engineers IEI Information Element Identifier IEIDL Information Element Identifier Data Length IETF Internet

Engineering Task Force IF Infrastructure IIoT Industrial Internet of Things IM Interference Measurement, Intermodulation, IP Multimedia IMC IMS Credentials IMEI International Mobile Equipment Identity IMGI International mobile group identity IMPI IP Multimedia Private Identity IMPU IP Multimedia PUBlic identity IMS IP Multimedia Subsystem IMSI International Mobile Subscriber Identity IoT Internet of Things IP Internet Protocol Isec IP Security, Internet Protocol Security IP-CAN IP-Connectivity Access Network IP-M IP Multicast IPv4 Internet Protocol Version 4 IPv6 Internet Protocol Version 6 IR Infrared IS In Sync IRP Integration Reference Point ISDN Integrated Services Digital Network ISIM IM Services Identity Module ISO International Organisation for Standardisation ISP Internet Service Provider IWF Interworking-Function I-WLAN Interworking WLAN Constraint length of the convolutional code, USIM Individual key kB Kilobyte (1000 bytes) kbps kilo-bits per second Kc Ciphering key Ki Individual subscriber authentication key KPI Key Performance Indicator KQI Key Quality Indicator KSI Key Set Identifier ksps kilo-symbols per second KVM Kernel Virtual Machine L1 Layer 1 (physical layer) L1-RSRP Layer 1 reference signal received power L2 Layer 2 (data link layer) L3 Layer 3 (network layer) LAA Licensed Assisted Access LAN Local Area Network LADN Local Area Data Network LBT Listen Before Talk LCM LifeCycle Management LCR Low Chip Rate LCS Location Services LCID Logical Channel ID LI Layer Indicator LLC Logical Link Control, Low Layer Compatibility LMF Location Management Function LOS Line of Sight LPLMN Local PLMN LPP LTE Positioning Protocol LSB Least Significant Bit LTE Long Term Evolution LWA LTE-WLAN aggregation LWIP LTE/WLAN Radio Level Integration with IPsec Tunnel LTE Long Term Evolution M2M Machine-to-Machine MAC Medium Access Control (protocol layering context) MAC Message authentication code (security/encryption context) MAC-A MAC used for authentication and key agreement (TSG T WG3 context) MAC-I MAC used for data integrity of signalling messages (TSG T WG3 context) MANO Management and Orchestration MBMS Multimedia Broadcast and Multicast Service MBSFN Multimedia Broadcast multicast service Single Frequency Network MCC Mobile Country Code MCG Master Cell Group MCOT Maximum Channel Occupancy Time MCS Modulation and coding scheme MDAF Management Data Analytics Function MDAS Management Data Analytics Service MDT Minimization of Drive Tests ME Mobile Equipment MeNB master eNB MER Message Error Ratio MGL Measurement Gap Length MGRP Measurement Gap Repetition Period MIB Master Information Block, Management Information Base MIMO Multiple Input Multiple Output MLC Mobile Location Centre MM Mobility Management MME Mobility Management Entity MN Master Node MNO Mobile Network Operator MO Measurement Object, Mobile Originated MPBCH MTC Physical Broadcast CHannel MPDCCH MTC Physical Downlink Control CHannel MPDSCH MTC Physical Downlink Shared CHannel MPRACH MTC Physical Random Access CHannel MPUSCH MTC Physical Uplink Shared Channel MPLS MultiProtocol Label Switching MS Mobile Station MSB Most Significant Bit MSC Mobile Switching Centre MSI Minimum System Information, MCH Scheduling Information MSID Mobile Station Identifier MSIN Mobile Station Identification Number MSISDN Mobile Subscriber ISDN Number MT Mobile Terminated, Mobile Termination MTC Machine-Type Communications mMTCmassive MTC, massive Machine-Type Communications MU-MIMO Multi User MIMO MWUS MTC wake-up signal, MTC WUS NACK Negative Acknowledgement NAI Network Access Identifier NAS Non-Access Stratum, Non-Access Stratum layer NCT Network Connectivity Topology NC-JT Non-Coherent Joint Transmission NEC Network Capability Exposure NE-DC NR-E-UTRA Dual Connectivity NEF Network Exposure Function NF Network Function NFP Network Forwarding Path NFPD Network Forwarding Path Descriptor NFV Network Functions Virtualization NFVI NFV Infrastructure NFVO NFV Orchestrator NG Next Generation, Next Gen NGEN-DC NG-RAN E-UTRA-NR Dual Connectivity NM Network Manager NMS Network Management System N-PoP Network Point of Presence NMIB, N-MIB Narrowband MIB NPBCH Narrowband Physical Broadcast CHannel NPDCCH Narrowband Physical Downlink Control CHannel NPDSCH Narrowband Physical

Download Shared Channel NPRACh Narrowband Primary Access Channel NPUSCH
Narrowband Physical Uplink Shared CHannel NPSS Narrowband Primary Synchronization Signal
NSSS Narrowband Secondary Synchronization Signal NR New Radio, Neighbour Relation NRF
NF Repository Function NRS Narrowband Reference Signal NS Network Service NSA Non-
Standalone operation mode NSD Network Service Descriptor NSR Network Service Record
NSSAI Network Slice Selection Assistance Information S-NNSAI Single-NSSAI NSSF Network
Slice Selection Function NW Network NWUS Narrowband wake-up signal, Narrowband WUS
NZP Non-Zero Power O&M Operation and Maintenance ODU2 Optical channel Data Unit - type 2
OFDM Orthogonal Frequency Division Multiplexing OFDMA Orthogonal Frequency Division
Multiple Access OOB Out-of-band OOS Out of Sync OPEX OPERating EXpense OSI Other
System Information OSS Operations Support System OTA over-the-air PAPR Peak-to-Average
Power Ratio PAR Peak to Average Ratio PBCH Physical Broadcast Channel PC Power Control,
Personal Computer PCC Primary Component Carrier, Primary CC P-CSCF Proxy CSCF PCell
Primary Cell PCI Physical Cell ID, Physical Cell Identity PCEF Policy and Charging Enforcement
Function PCF Policy Control Function PCRF Policy Control and Charging Rules Function PDCP
Packet Data Convergence Protocol, Packet Data Convergence Protocol layer PDCCH Physical
Downlink Control Channel PDCP Packet Data Convergence Protocol PDN Packet Data Network,
Public Data Network PDSCH Physical Downlink Shared Channel PDU Protocol Data Unit PEI
Permanent Equipment Identifiers PFD Packet Flow Description P-GW PDN Gateway PHICH
Physical hybrid-ARQ indicator channel PHY Physical layer PLMN Public Land Mobile Network
PIN Personal Identification Number PM Performance Measurement PMI Precoding Matrix
Indicator PNF Physical Network Function PNFD Physical Network Function Descriptor PNFR
Physical Network Function Record POC PTT over Cellular PP, PTP Point-to-Point PPP Point-to-
Point Protocol PRACH Physical RACH PRB Physical resource block PRG Physical resource block
group ProSe Proximity Services, Proximity-Based Service PRS Positioning Reference Signal PRR
Packet Reception Radio PS Packet Services PSBCH Physical Sidelink Broadcast Channel PSDCH
Physical Sidelink Downlink Channel PSCCH Physical Sidelink Control Channel PSSCH Physical
Sidelink Shared Channel PSCell Primary SCell PSS Primary Synchronization Signal PSTN Public
Switched Telephone Network PT-RS Phase-tracking reference signal PTT Push-to-Talk PUCCH
Physical Uplink Control Channel PUSCH Physical Uplink Shared Channel QAM Quadrature
Amplitude Modulation QCI QoS class of identifier QCL Quasi co-location QFI QoS Flow ID, QoS
Flow Identifier QoS Quality of Service QPSK Quadrature (Quaternary) Phase Shift Keying QZSS
Quasi-Zenith Satellite System RA-RNTI Random Access RNTI RAB Radio Access Bearer,
Random Access Burst RACH Random Access Channel RADIUS Remote Authentication Dial In
User Service RAN Radio Access Network RAND RANDom number (used for authentication)
RAR Random Access Response RAT Radio Access Technology RAU Routing Area Update RB
Resource block, Radio Bearer RBG Resource block group REG Resource Element Group Rel
Release REQ REQuest RF Radio Frequency RI Rank Indicator RIV Resource indicator value RL
Radio Link RLC Radio Link Control, Radio Link Control layer RLC AM RLC Acknowledged
Mode RLC UM RLC Unacknowledged Mode RLF Radio Link Failure RLM Radio Link
Monitoring RLM-RS Reference Signal for RLM RM Registration Management RMC Reference
Measurement Channel RMSI Remaining MSI, Remaining Minimum System Information RN
Relay Node RNC Radio Network Controller RNL Radio Network Layer RNTI Radio Network
Temporary Identifier ROHC RObust Header Compression RRC Radio Resource Control, Radio
Resource Control layer RRM Radio Resource Management RS Reference Signal RSRP Reference
Signal Received Power RSRQ Reference Signal Received Quality RSSI Received Signal Strength
Indicator RSU Road Side Unit RSTD Reference Signal Time difference RTP Real Time Protocol
RTS Ready-To-Send RTT Round Trip Time Rx Reception, Receiving, Receiver S1AP S1
Application Protocol S1-MME S1 for the control plane S1-U S1 for the user plane S-CSCF serving
CSCF S-GW Serving Gateway S-RNTI SRNC Radio Network Temporary Identity S-TMSI SAE

Temporary Mobile Station Identifier SA Standalone operation mode SAE System Architecture Evolution SAP Service Access Point SAPD Service Access Point Descriptor SAPI Service Access Point Identifier SCC Secondary Component Carrier, Secondary CC SCell Secondary Cell SCEF Service Capability Exposure Function SC-FDMA Single Carrier Frequency Division Multiple Access SCG Secondary Cell Group SCM Security Context Management SCS Subcarrier Spacing SCTP Stream Control Transmission Protocol SDAP Service Data Adaptation Protocol, Service Data Adaptation Protocol layer SDL Supplementary Downlink SDNF Structured Data Storage Network Function SDP Session Description Protocol SDSF Structured Data Storage Function SDT Small Data Transmission SDU Service Data Unit SEAF Security Anchor Function SeNB secondary eNB SEPP Security Edge Protection Proxy SFI Slot format indication SFTD Space-Frequency Time Diversity, SFN and frame timing difference SFN System Frame Number SgNB Secondary gNB SGSN Serving GPRS Support Node S-GW Serving Gateway SI System Information SI-RNTI System Information RNTI SIB System Information Block SIM Subscriber Identity Module SIP Session Initiated Protocol SiP System in Package SL Sidelink SLA Service Level Agreement SM Session Management SMF Session Management Function SMS Short Message Service SMSF SMS Function SMTC SSB-based Measurement Timing Configuration SN Secondary Node, Sequence Number SoC System on Chip SON Self-Organizing Network SpCell Special Cell SP-CSI-RNTI Semi-Persistent CSI RNTI SPS Semi-Persistent Scheduling SQN Sequence number SR Scheduling Request SRB Signalling Radio Bearer SRS Sounding Reference Signal SS Synchronization Signal SSB Synchronization Signal Block SSID Service Set Identifier SS/PBCH SS/PBCH Block Resource Indicator, Synchronization Block SSBRI Signal Block Resource Indicator SSC Session and Service Continuity SS-RSRP Synchronization Signal based Reference Signal Received Power SS-RSRQ Synchronization Signal based Reference Signal Received Quality SS-SINR Synchronization Signal based Signal to Noise and Interference Ratio SSS Secondary Synchronization Signal SSSG Search Space Set Group SSSIF Search Space Set Indicator SST Slice/Service Types SU-MIMO Single User MIMO SUL Supplementary Uplink TA Timing Advance, Tracking Area TAC Tracking Area Code TAG Timing Advance Group TAI Tracking Area Identity TAU Tracking Area Update TB Transport Block TBS Transport Block Size TBD To Be Defined TCI Transmission Configuration Indicator TCP Transmission Communication Protocol TDD Time Division Duplex TDM Time Division Multiplexing TDMA Time Division Multiple Access TE Terminal Equipment TEID Tunnel End Point Identifier TFT Traffic Flow Template TMSI Temporary Mobile Subscriber Identity TNL Transport Network Layer TPC Transmit Power Control TPMI Transmitted Precoding Matrix Indicator TR Technical Report TRP, TRxP Transmission Reception Point TRS Tracking Reference Signal TRx Transceiver TS Technical Specifications, Technical Standard TTI Transmission Time Interval Tx Transmission, Transmitting, Transmitter U-RNTI UTRAN Radio Network Temporary Identity UART Universal Asynchronous Receiver and Transmitter UCI Uplink Control Information UE User Equipment UDM Unified Data Management UDP User Datagram Protocol UDSF Unstructured Data Storage Network Function UICC Universal Integrated Circuit Card UL Uplink UM Unacknowledged Mode UML Unified Modelling Language UMTS Universal Mobile Telecommunications System UP User Plane UPF User Plane Function URI Uniform Resource Identifier URL Uniform Resource Locator URLLC Ultra-Reliable and Low Latency USB Universal Serial Bus USIM Universal Subscriber Identity Module USS UE-specific search space UTRA UMTS Terrestrial Radio Access UTRAN Universal Terrestrial Radio Access Network UwPTS Uplink Pilot Time Slot V2I Vehicle-to-Infrastructure V2P Vehicle-to-Pedestrian V2V Vehicle-to-Vehicle V2X Vehicle-to-everything VIM Virtualized Infrastructure Manager VL Virtual Link, VLAN Virtual LAN, Virtual Local Area Network VM Virtual Machine VNF Virtualized Network Function VNFFG VNF Forwarding Graph VNFFGD VNF Forwarding Graph Descriptor VNFM VNF Manager VoIP Voice-over-IP, Voice-over-Internet Protocol VPLMN Visited Public Land Mobile Network VPN Virtual Private Network VRB Virtual Resource Block WiMAX Worldwide Interoperability for Microwave Access

WLAN Wireless Local Area Network WMAN Wireless Metropolitan Area Network WPAN
Wireless Personal Area Network X2-C X2-Control plane X2-U X2-User plane XML eXtensible
Markup Language XRES EXpected user RESponse XOR eXclusive OR ZC Zadoff-Chu ZP Zero
Power

Terminology

(186) For the purposes of the present document, the following terms and definitions are applicable to the examples and embodiments discussed herein.

(187) The term “circuitry” as used herein refers to, is part of, or includes hardware components such as an electronic circuit, a logic circuit, a processor (shared, dedicated, or group) and/or memory (shared, dedicated, or group), an Application Specific Integrated Circuit (ASIC), a field-programmable device (FPD) (e.g., a field-programmable gate array (FPGA), a programmable logic device (PLD), a complex PLD (CPLD), a high-capacity PLD (HCPLD), a structured ASIC, or a programmable SoC), digital signal processors (DSPs), etc., that are configured to provide the described functionality. In some embodiments, the circuitry may execute one or more software or firmware programs to provide at least some of the described functionality. The term “circuitry” may also refer to a combination of one or more hardware elements (or a combination of circuits used in an electrical or electronic system) with the program code used to carry out the functionality of that program code. In these embodiments, the combination of hardware elements and program code may be referred to as a particular type of circuitry.

(188) The term “processor circuitry.” as used herein refers to, is part of, or includes circuitry capable of sequentially and automatically carrying out a sequence of arithmetic or logical operations, or recording, storing, and/or transferring digital data. Processing circuitry may include one or more processing cores to execute instructions and one or more memory structures to store program and data information. The term “processor circuitry” may refer to one or more application processors, one or more baseband processors, a physical central processing unit (CPU), a single-core processor, a dual-core processor, a triple-core processor, a quad-core processor, and/or any other device capable of executing or otherwise operating computer-executable instructions, such as program code, software modules, and/or functional processes. Processing circuitry may include more hardware accelerators, which may be microprocessors, programmable processing devices, or the like. The one or more hardware accelerators may include, for example, computer vision (CV) and/or deep learning (DL) accelerators. The terms “application circuitry” and/or “baseband circuitry” may be considered synonymous to, and may be referred to as, “processor circuitry.”

(189) The term “interface circuitry” as used herein refers to, is part of, or includes circuitry that enables the exchange of information between two or more components or devices. The term “interface circuitry” may refer to one or more hardware interfaces, for example, buses, I/O interfaces, peripheral component interfaces, network interface cards, and/or the like.

(190) The term “user equipment” or “UE” as used herein refers to a device with radio communication capabilities and may describe a remote user of network resources in a communications network. The term “user equipment” or “UE” may be considered synonymous to, and may be referred to as, client, mobile, mobile device, mobile terminal, user terminal, mobile unit, mobile station, mobile user, subscriber, user, remote station, access agent, user agent, receiver, radio equipment, reconfigurable radio equipment, reconfigurable mobile device, etc. Furthermore, the term “user equipment” or “UE” may include any type of wireless/wired device or any computing device including a wireless communications interface.

(191) The term “network element” as used herein refers to physical or virtualized equipment and/or infrastructure used to provide wired or wireless communication network services. The term “network element” may be considered synonymous to and/or referred to as a networked computer, networking hardware, network equipment, network node, router, switch, hub, bridge, radio network controller, RAN device, RAN node, gateway, server, virtualized VNF, NFVI, and/or the like.

(192) The term “computer system” as used herein refers to any type interconnected electronic

devices, computer devices, or components thereof. Additionally, the term “computer system” and/or “system” may refer to various components of a computer that are communicatively coupled with one another. Furthermore, the term “computer system” and/or “system” may refer to multiple computer devices and/or multiple computing systems that are communicatively coupled with one another and configured to share computing and/or networking resources.

(193) The term “appliance,” “computer appliance,” or the like, as used herein refers to a computer device or computer system with program code (e.g., software or firmware) that is specifically designed to provide a specific computing resource. A “virtual appliance” is a virtual machine image to be implemented by a hypervisor-equipped device that virtualizes or emulates a computer appliance or otherwise is dedicated to provide a specific computing resource.

(194) The term “resource” as used herein refers to a physical or virtual device, a physical or virtual component within a computing environment, and/or a physical or virtual component within a particular device, such as computer devices, mechanical devices, memory space, processor/CPU time, processor/CPU usage, processor and accelerator loads, hardware time or usage, electrical power, input/output operations, ports or network sockets, channel/link allocation, throughput, memory usage, storage, network, database and applications, workload units, and/or the like. A “hardware resource” may refer to compute, storage, and/or network resources provided by physical hardware element(s). A “virtualized resource” may refer to compute, storage, and/or network resources provided by virtualization infrastructure to an application, device, system, etc. The term “network resource” or “communication resource” may refer to resources that are accessible by computer devices/systems via a communications network. The term “system resources” may refer to any kind of shared entities to provide services, and may include computing and/or network resources. System resources may be considered as a set of coherent functions, network data objects or services, accessible through a server where such system resources reside on a single host or multiple hosts and are clearly identifiable.

(195) The term “channel” as used herein refers to any transmission medium, either tangible or intangible, which is used to communicate data or a data stream. The term “channel” may be synonymous with and/or equivalent to “communications channel,” “data communications channel,” “transmission channel,” “data transmission channel,” “access channel,” “data access channel,” “link,” “data link,” “carrier,” “radiofrequency carrier,” and/or any other like term denoting a pathway or medium through which data is communicated. Additionally, the term “link” as used herein refers to a connection between two devices through a RAT for the purpose of transmitting and receiving information.

(196) The terms “instantiate,” “instantiation,” and the like as used herein refers to the creation of an instance. An “instance” also refers to a concrete occurrence of an object, which may occur, for example, during execution of program code.

(197) The terms “coupled,” “communicatively coupled,” along with derivatives thereof are used herein. The term “coupled” may mean two or more elements are in direct physical or electrical contact with one another, may mean that two or more elements indirectly contact each other but still cooperate or interact with each other, and/or may mean that one or more other elements are coupled or connected between the elements that are said to be coupled with each other. The term “directly coupled” may mean that two or more elements are in direct contact with one another. The term “communicatively coupled” may mean that two or more elements may be in contact with one another by a means of communication including through a wire or other interconnect connection, through a wireless communication channel or link, and/or the like.

(198) The term “information element” refers to a structural element containing one or more fields. The term “field” refers to individual contents of an information element, or a data element that contains content.

(199) The term “SMTC” refers to an SSB-based measurement timing configuration configured by SSB-MeasurementTimingConfiguration.

(200) The term “SSB” refers to an SS/PBCH block.

(201) The term “a “Primary Cell” refers to the MCG cell, operating on the primary frequency, in which the UE either performs the initial connection establishment procedure or initiates the connection re-establishment procedure.

(202) The term “Primary SCG Cell” refers to the SCG cell in which the UE performs random access when performing the Reconfiguration with Sync procedure for DC operation.

(203) The term “Secondary Cell” refers to a cell providing additional radio resources on top of a Special Cell for a UE configured with CA.

(204) The term “Secondary Cell Group” refers to the subset of serving cells comprising the PSCell and zero or more secondary cells for a UE configured with DC.

(205) The term “Serving Cell” refers to the primary cell for a UE in RRC_CONNECTED not configured with CA/DC there is only one serving cell comprising of the primary cell.

(206) The term “serving cell” or “serving cells” refers to the set of cells comprising the Special Cell(s) and all secondary cells for a UE in RRC_CONNECTED configured with CA/.

(207) The term “Special Cell” refers to the PCell of the MCG or the PSCell of the SCG for DC operation: otherwise, the term “Special Cell” refers to the Pcell.

(208) The term “application” may refer to a complete and deployable package, environment to achieve a certain function in an operational environment. The term “AI/ML application” or the like may be an application that contains some AI/ML models and application-level descriptions.

(209) The term “machine learning” or “ML” refers to the use of computer systems implementing algorithms and/or statistical models to perform specific task(s) without using explicit instructions, but instead relying on patterns and inferences. ML algorithms build or estimate mathematical model(s) (referred to as “ML models” or the like) based on sample data (referred to as “training data.” “model training information,” or the like) in order to make predictions or decisions without being explicitly programmed to perform such tasks. Generally, an ML algorithm is a computer program that learns from experience with respect to some task and some performance measure, and an ML model may be any object or data structure created after an ML algorithm is trained with one or more training datasets. After training, an ML model may be used to make predictions on new datasets. Although the term “ML algorithm” refers to different concepts than the term “ML model,” these terms as discussed herein may be used interchangeably for the purposes of the present disclosure.

(210) The term “machine learning model.” “ML model,” or the like may also refer to ML methods and concepts used by an ML-assisted solution. An “ML-assisted solution” is a solution that addresses a specific use case using ML algorithms during operation. ML models include supervised learning (e.g., linear regression, k-nearest neighbor (KNN), decision tree algorithms, support machine vectors. Bayesian algorithm, ensemble algorithms, etc.) unsupervised learning (e.g., K-means clustering, principle component analysis (PCA), etc.), reinforcement learning (e.g., Q-learning, multi-armed bandit learning, deep RL, etc.), neural networks, and the like. Depending on the implementation a specific ML model could have many sub-models as components and the ML model may train all sub-models together. Separately trained ML models can also be chained together in an ML pipeline during inference. An “ML pipeline” is a set of functionalities, functions, or functional entities specific for an ML-assisted solution; an ML pipeline may include one or several data sources in a data pipeline, a model training pipeline, a model evaluation pipeline, and an actor. The “actor” is an entity that hosts an ML assisted solution using the output of the ML model inference). The term “ML training host” refers to an entity, such as a network function, that hosts the training of the model. The term “ML inference host” refers to an entity, such as a network function, that hosts model during inference mode (which includes both the model execution as well as any online learning if applicable). The ML-host informs the actor about the output of the ML algorithm, and the actor takes a decision for an action (an “action” is performed by an actor as a result of the output of an ML assisted solution). The term “model inference information” refers to

information used as an input to the ML model for determining inference(s): the data used to train an ML model and the data used to determine inferences may overlap, however, “training data” and “inference data” refer to different concepts.

Claims

1. An apparatus comprising: memory to store policy statement information for a plurality of radio access network (RAN) automation applications (rApps); and processor circuitry, coupled with the memory, to: retrieve the policy statement information from the memory, wherein the policy statement information includes respective policy scope identifiers for respective rApps in the plurality of rApps; identify a conflict associated with common or overlapping policy scope identifiers between two or more rApps from the plurality of rApps; modify one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict; and notify the two or more rApps of the modification of the one or more A1 policies.
2. The apparatus of claim 1, wherein the processor circuitry is further to: track an enforcement status of an A1 policy created in the non-RT RIC; and notify an rApp subscribed to the tracked A1 policy when the enforcement status of the tracked A1 policy changes.
3. The apparatus of claim 1, wherein the processor circuitry is further to track a supported A1 policy type for each of a plurality of near-RT RICs.
4. The apparatus of claim 3, wherein the processor circuitry is to notify an rApp of a supported A1 policy type for a near-RT RIC based on a policy identifier.
5. The apparatus of claim 3, wherein the processor circuitry is to notify an rApp of a change in supported policy types for a near-RT RIC.
6. The apparatus of claim 1, wherein the processor circuitry is further to generate an A1-related service that includes: a RAN intent discovery service, an A1 policy management service, or an A1 policy type discovery service.
7. The apparatus of claim 1, wherein the processor circuitry is further to: receive a supported A1 policy types query request via an R1 termination; and in response to the supported A1 policy types query request, send a query response that includes an indication of supported A1 policy types to a policy generation rApp via the R1 termination.
8. The apparatus of claim 1, wherein the processor circuitry is further to: receive an A1 policy feedback message from the near-RT RIC via an A1 termination to indicate a change of A1 policy enforcement status; and send a notification of the A1 policy enforcement status change to a policy generation rApp via an R1 termination.
9. The apparatus of claim 1, wherein the processor circuitry is further to: receive an A1 policy modification request from a policy generation rApp via an R1 termination; update an A1 policy over the A1 interface based on the A1 policy modification request; and send an A1 policy modification response to the policy generation rApp via the R1 termination.
10. The apparatus of claim 1, wherein the processor circuitry is further to: receive an A1 policy deletion request from a policy generation rApp via an R1 termination; remove an A1 policy over the A1 interface based on the A1 policy deletion request; and send an A1 policy deletion response to the policy generation rApp via the R1 termination.
11. One or more non-transitory computer-readable media storing instructions that, when executed by one or more processors, cause an A1 policy function to: identify a conflict associated with common or overlapping policy scope identifiers between two or more radio access network (RAN) automation applications (rApps) from a plurality of rApps based on policy statement information that includes respective policy scope identifiers for respective rApps in the plurality of rApps; modify one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict;

and notify the two or more rApps of the modification of the one or more A1 policies.

12. The one or more non-transitory computer-readable media of claim 11, wherein the media further stores instructions to: track an enforcement status of an A1 policy created in the non-RT RIC; and notify an rApp subscribed to the tracked A1 policy when the enforcement status of the tracked A1 policy changes.

13. The one or more non-transitory computer-readable media of claim 11, wherein the media further stores instructions to: track a supported A1 policy type for each of a plurality of near-RT RICs; and notify an rApp of a supported A1 policy type for a near-RT RIC based on a policy identifier; or notify an rApp of a change in supported policy types for a near-RT RIC.

14. The one or more non-transitory computer-readable media of claim 11, wherein the media further stores instructions to: receive a supported A1 policy types query request via an R1 termination; and in response to the supported A1 policy types query request, send a query response that includes an indication of supported A1 policy types to a policy generation rApp via the R1 termination.

15. The one or more non-transitory computer-readable media of claim 11, wherein the media further stores instructions to: receive an A1 policy feedback message from the near-RT RIC via an A1 termination to indicate a change of A1 policy enforcement status; and send a notification of the A1 policy enforcement status change to a policy generation rApp via an R1 termination.

16. The one or more non-transitory computer-readable media of claim 11, wherein the media further stores instructions to: receive an A1 policy modification request from a policy generation rApp via an R1 termination; update an A1 policy over the A1 interface based on the A1 policy modification request; and send an A1 policy modification response to the policy generation rApp via the R1 termination.

17. The one or more non-transitory computer-readable media of claim 11, wherein the media further stores instructions to generate an A1-related service that includes: a RAN intent discovery service, an A1 policy management service, or an A1 policy type discovery service.

18. One or more non-transitory computer-readable media storing instructions that, when executed by one or more processors, cause an A1 policy function to: identify a conflict associated with common or overlapping policy scope identifiers between two or more radio access network (RAN) automation applications (rApps) from a plurality of rApps based on policy statement information that includes respective policy scope identifiers for respective rApps in the plurality of rApps; modify one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict; notify the two or more rApps of the modification of the one or more A1 policies; and track a supported A1 policy type for each of a plurality of near-RT RICs, and either: notify an rApp of a supported A1 policy type for a near-RT RIC based on a policy identifier, or notify an rApp of a change in supported policy types for a near-RT RIC.

19. The one or more non-transitory computer-readable media of claim 18, wherein the media further stores instructions to send, to an rApp via an R1 termination, one or more of; an indication of a change in an enforcement status of a tracked A1 policy; an indication of supported A1 policy types; and a notification of an A1 policy enforcement status change.

20. The one or more non-transitory computer-readable media of claim 18, wherein the media further stores instructions to generate an A1-related service that includes: a RAN intent discovery service, an A1 policy management service, or an A1 policy type discovery service.
